

Daneshkar Presentation
for
Data Analysis
of
Tennis Games

Partipants:

Marzie Dezfoulianrad

Seyyed Mohammad Osuli Dezfuli Zade

Mozhgan Mohammadi

We'll analyze tennis game data by exploring 17 key questions from a large dataset of matches. Our presentation will highlight the insights we've uncovered from this rich information.

Question 1:

How many tennis players are included in the dataset?

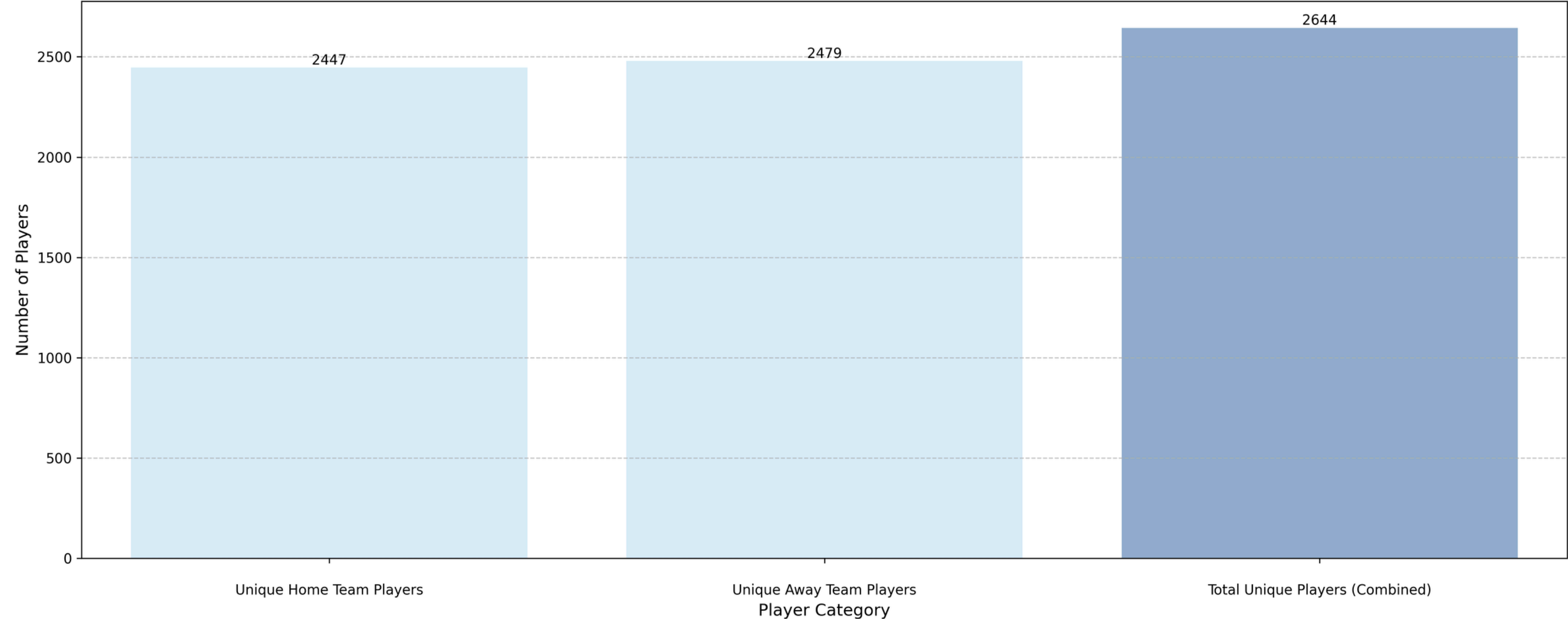
Number of Tennis Players in the Dataset (Excluding Nulls)

Unique Home Team Players 2447

Unique Away Team Players 2479

Total Unique Players (Combined) 2644

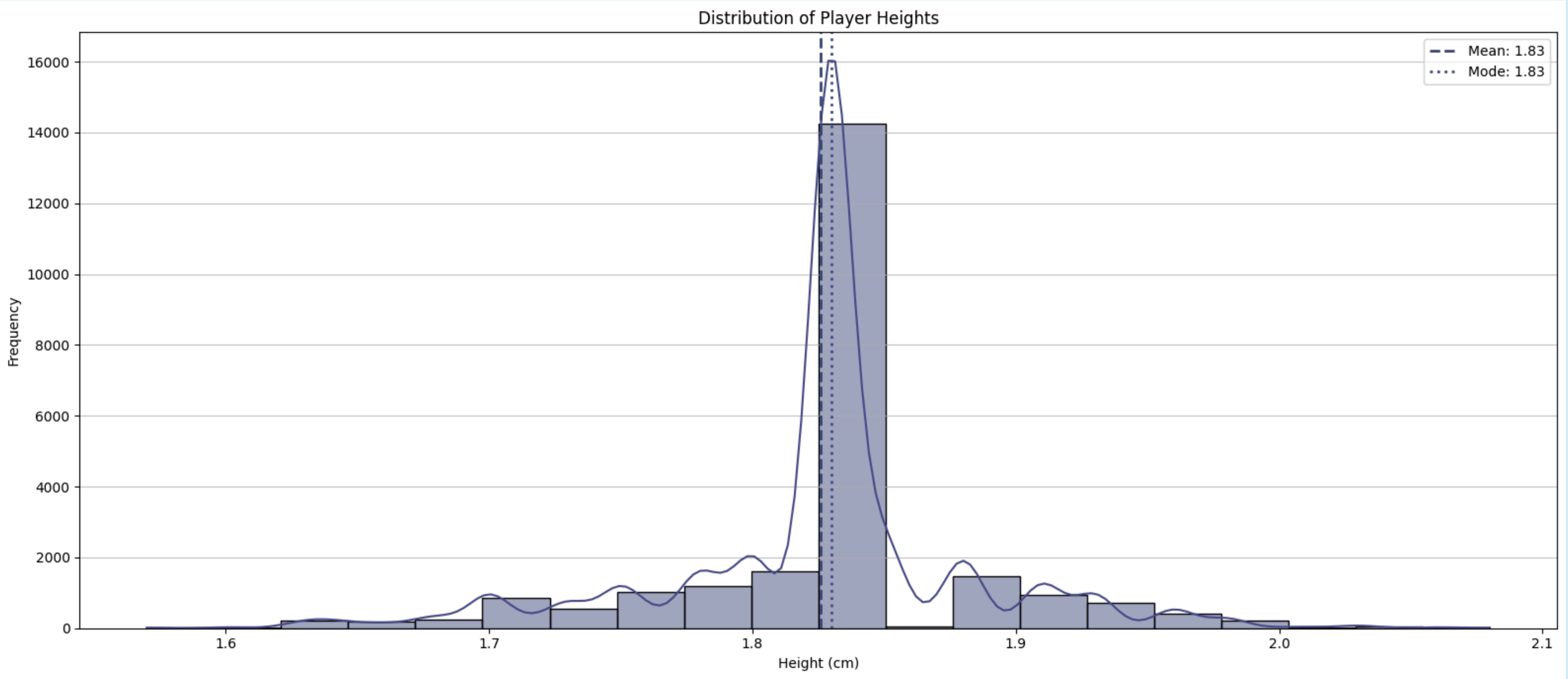
Number of Unique Tennis Players by Category



Question 2 :

What is the average height of the players?

Combined all players data shape:
Median height used for filling NaN values: 1.83
The Arithmetic Mean of Player Heights is: 1.83
Mode of Height: 1.83

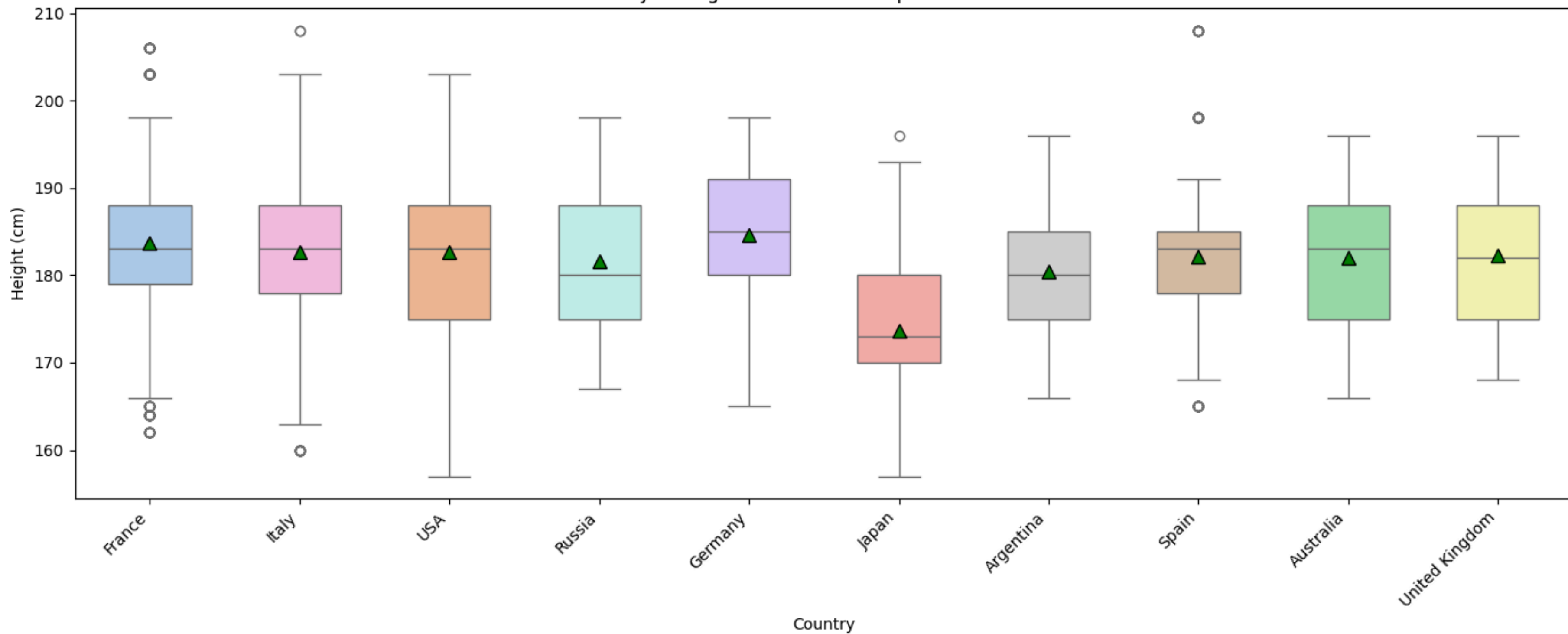


**We have a tall, narrow bell curve peaking at
1.83 cm.**

**About 95% of all players fall between 1.78 cm
and 1.88 cm.**

**There is a light left skew, which shows there
are more players below 1.83 cm than above it
and if we look closely we see that the tail
stretches slightly more to the left than to the
right.**

Player Height Distribution - Top-10 Countries



Question 3 :

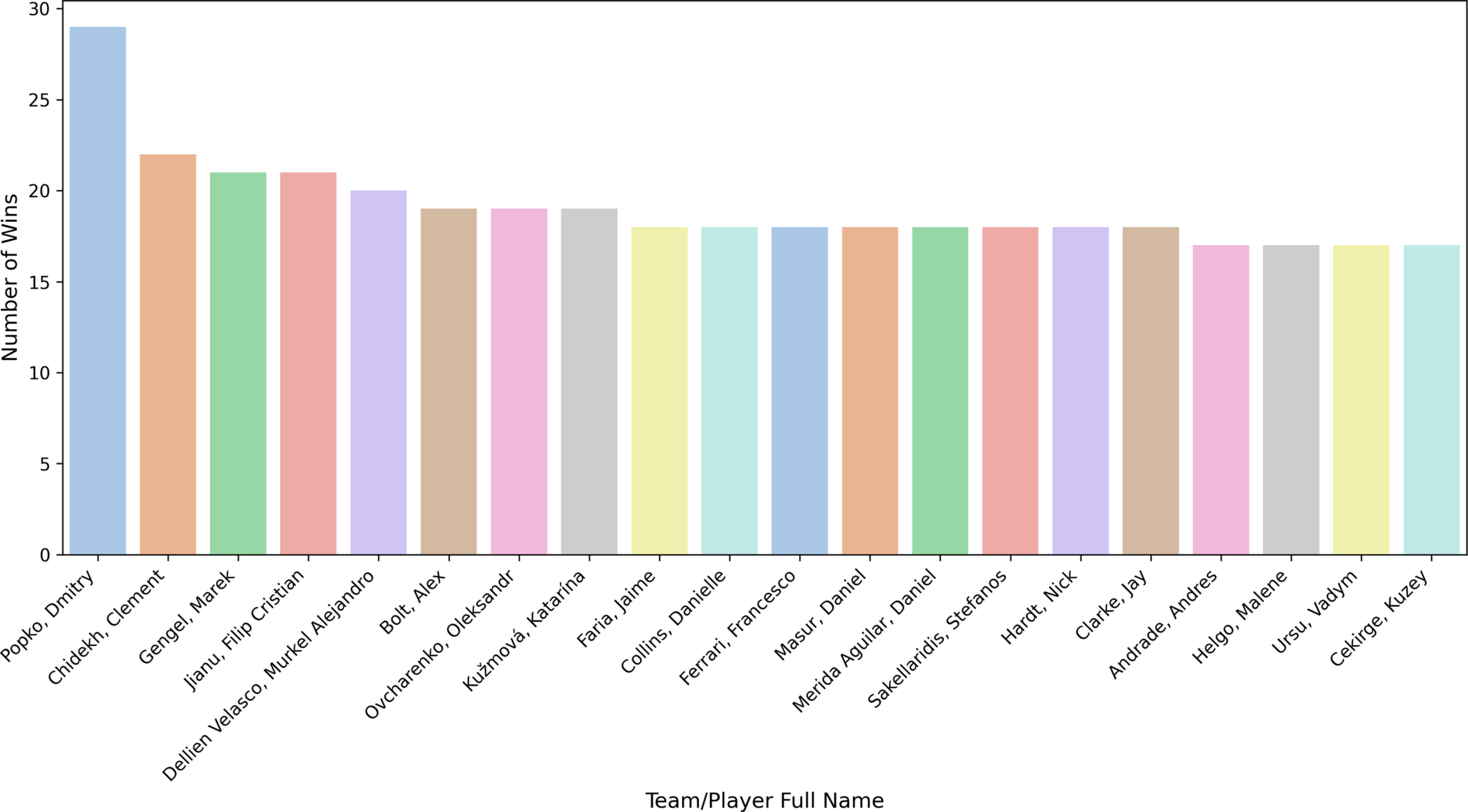
Which player has the highest number of wins?

Total Actual Winners Found : 2320

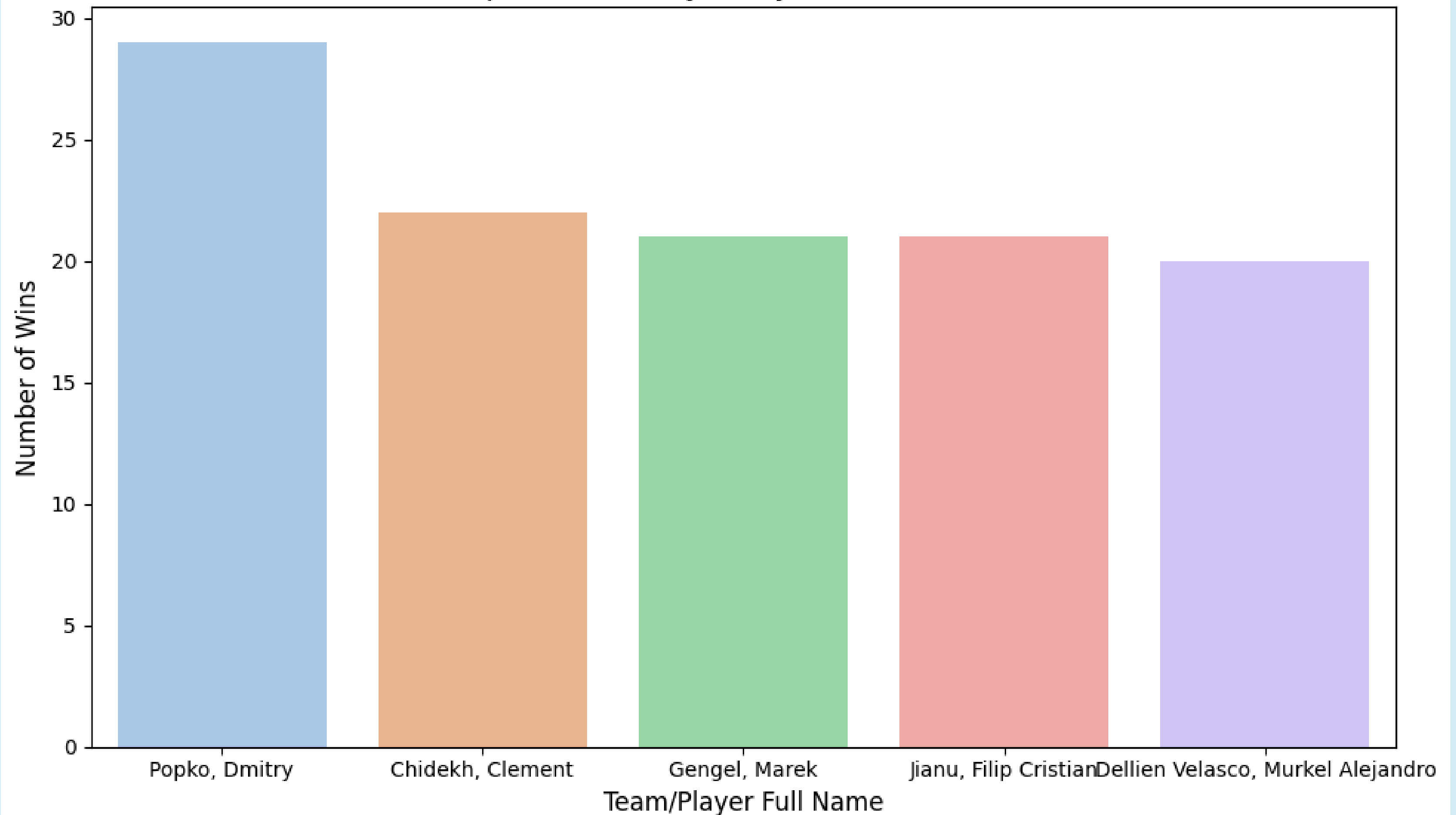
Winning Full Name

Popko, Dmitry	29
Chidekh, Clement	22
Gengel, Marek	21
Jianu, Filip Cristian	21
Dellien Velsco, Murkel Alejandro	20

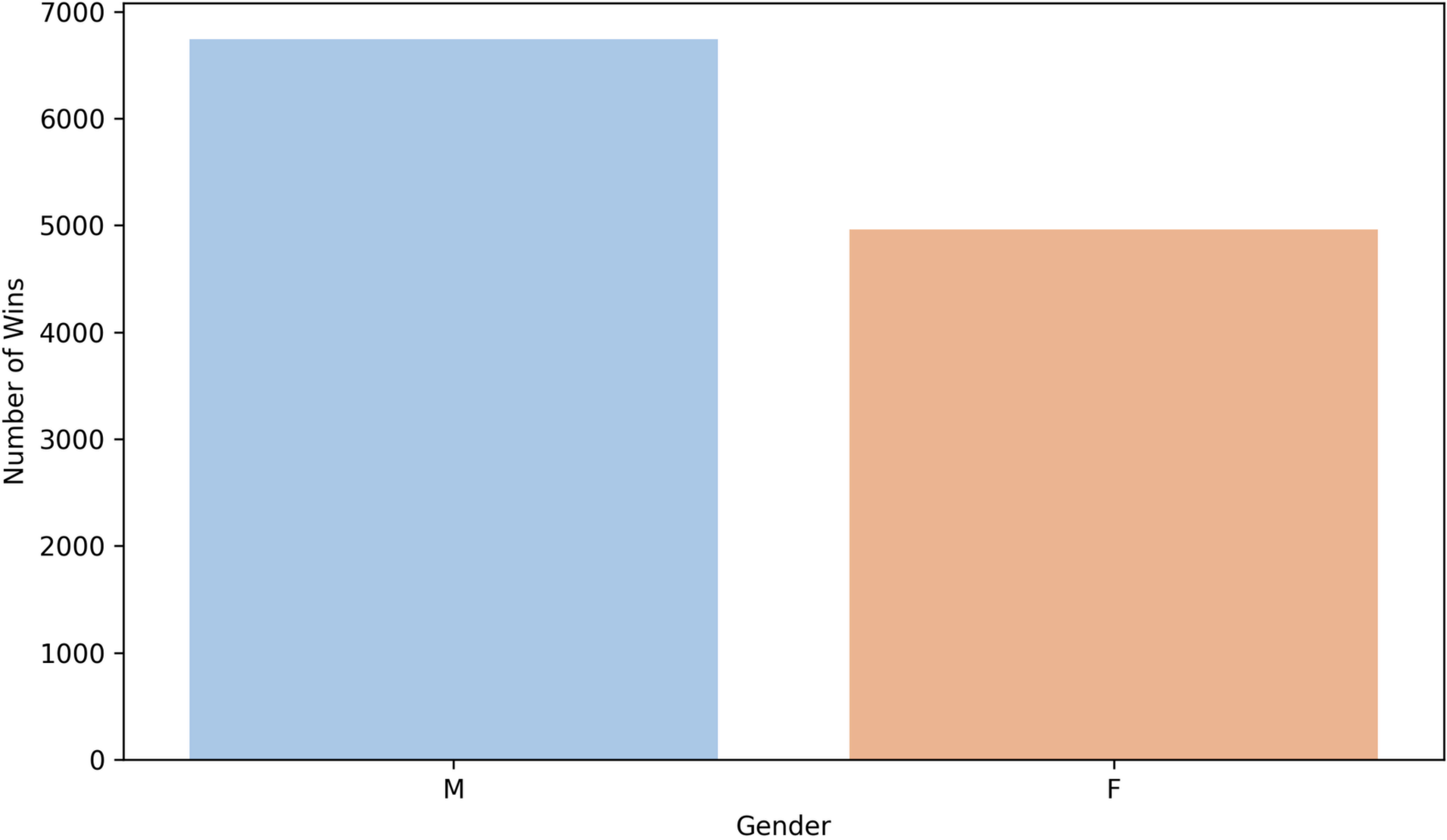
Top 20 Teams/Players by Number of Wins



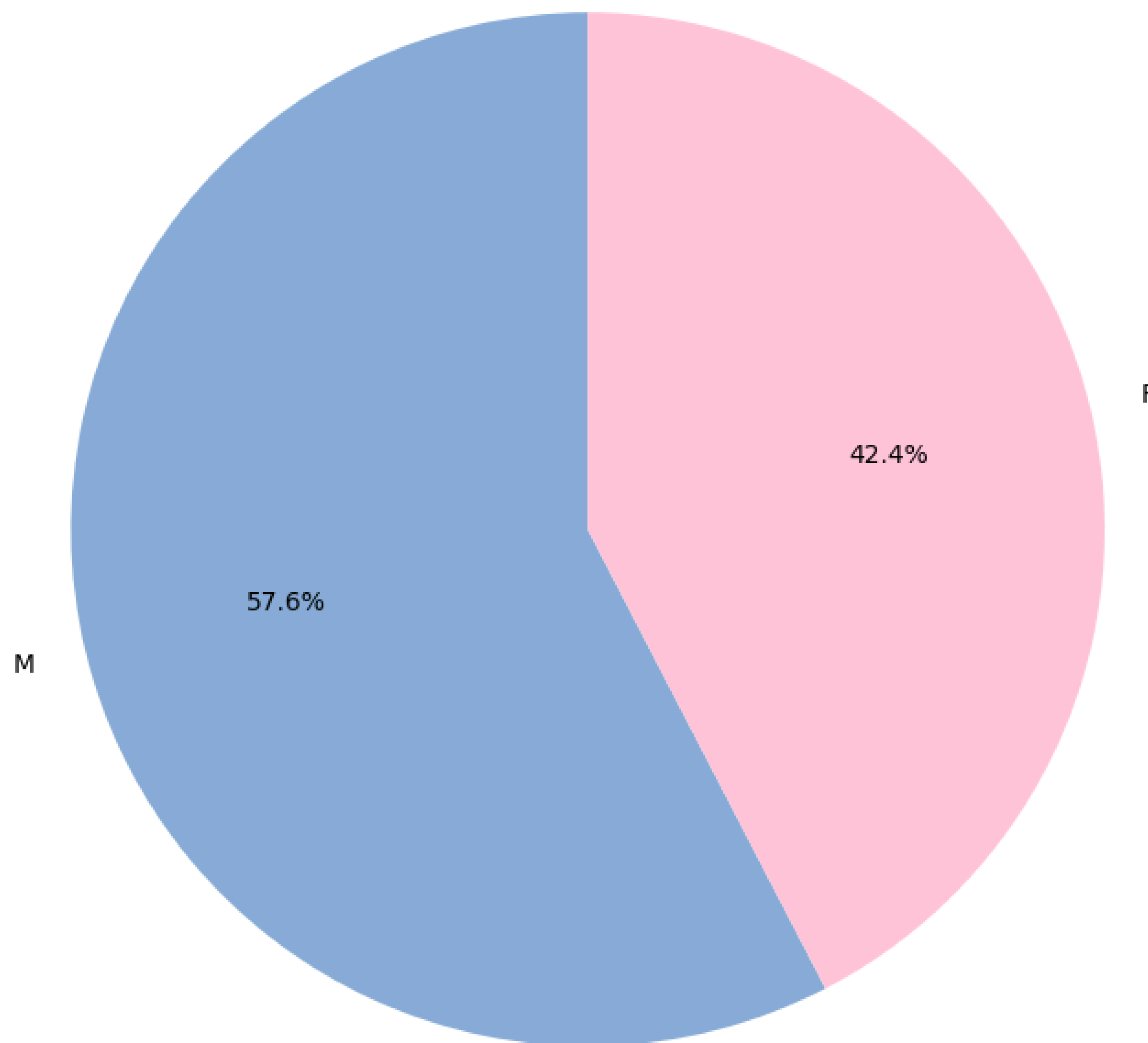
Top 5 Teams/Players by Number of Wins



Number of Wins by Gender



Percentage of Wins by Gender



Question 4 :

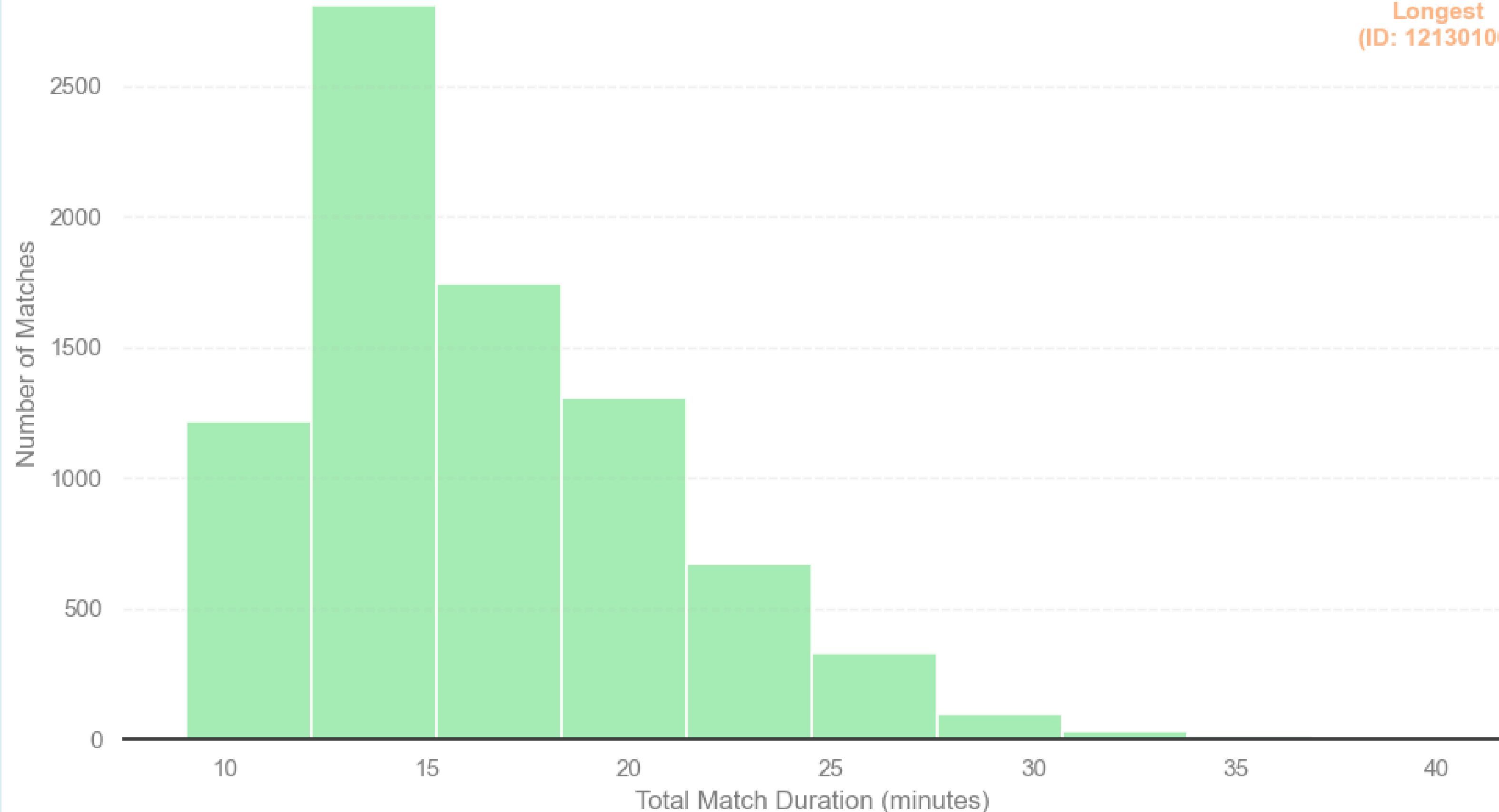
What is the longest match recorded in terms of duration?

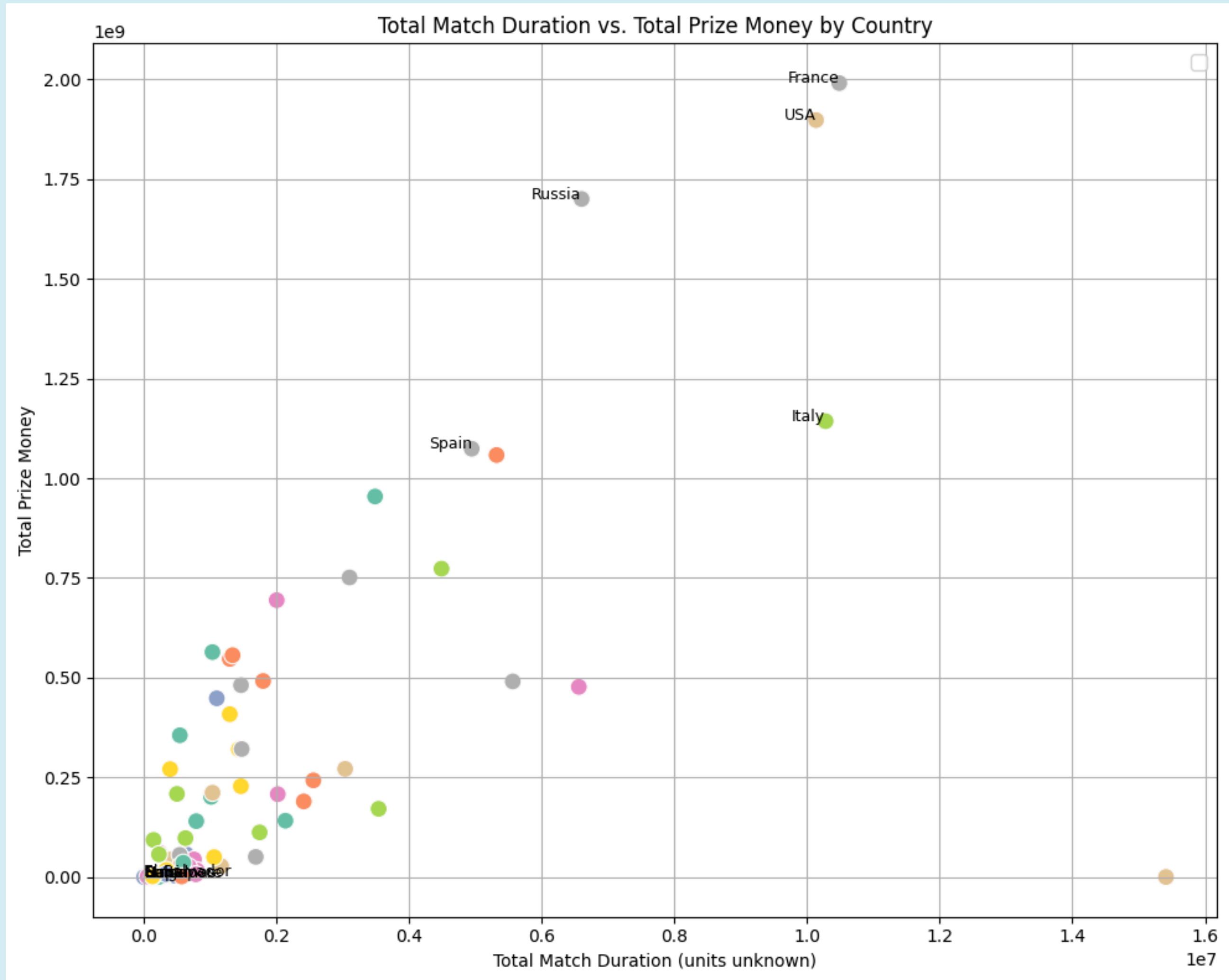
ID of the Longest Match: 12151527

Duration: 0h 15m 31s

Distribution of Tennis Match Durations

Longest
(ID: 12130100)



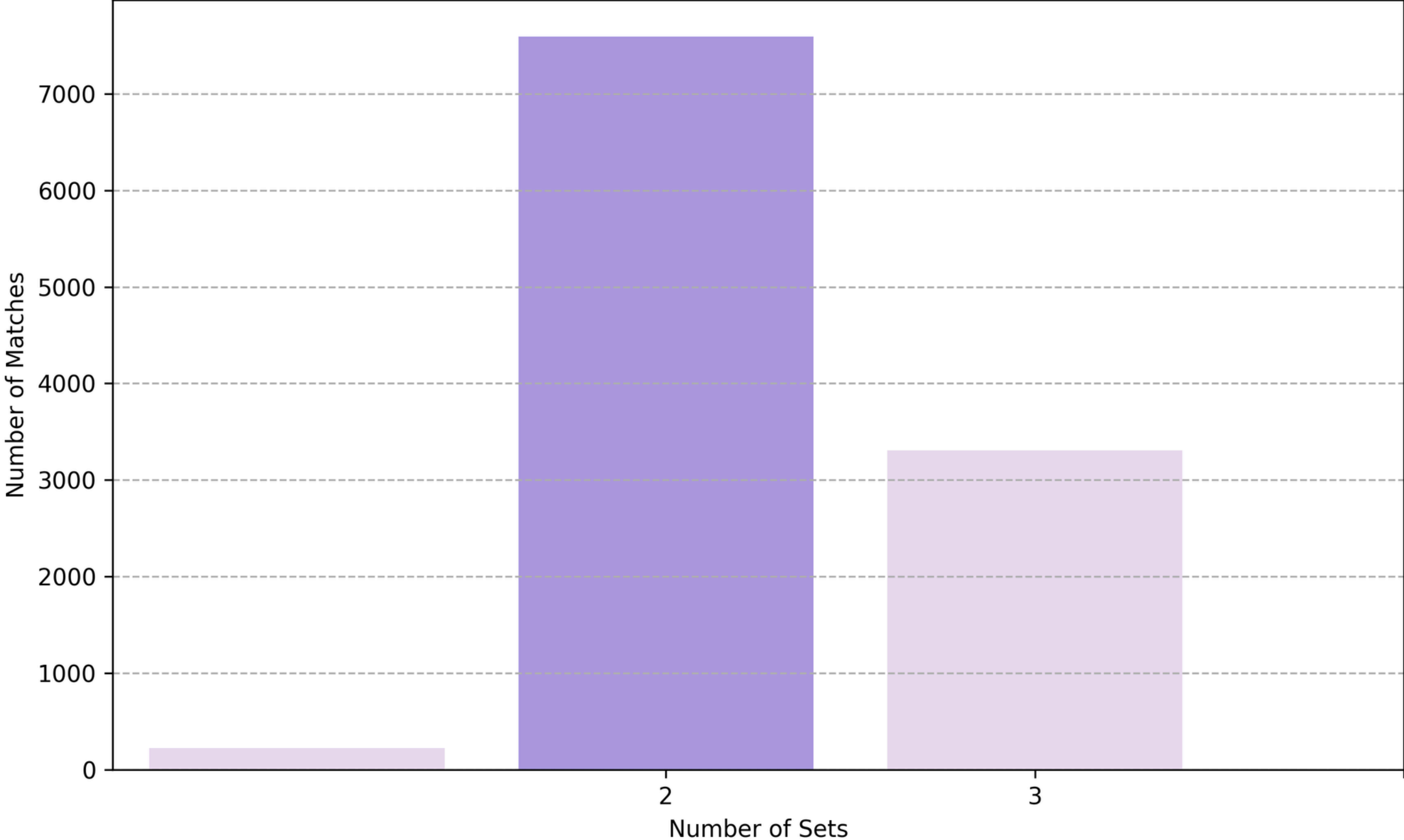


Question 5 :

How many sets are typically played in a tennis match?

The average number of sets played per match is: 2.28

Distribution of Total Sets Played per Match

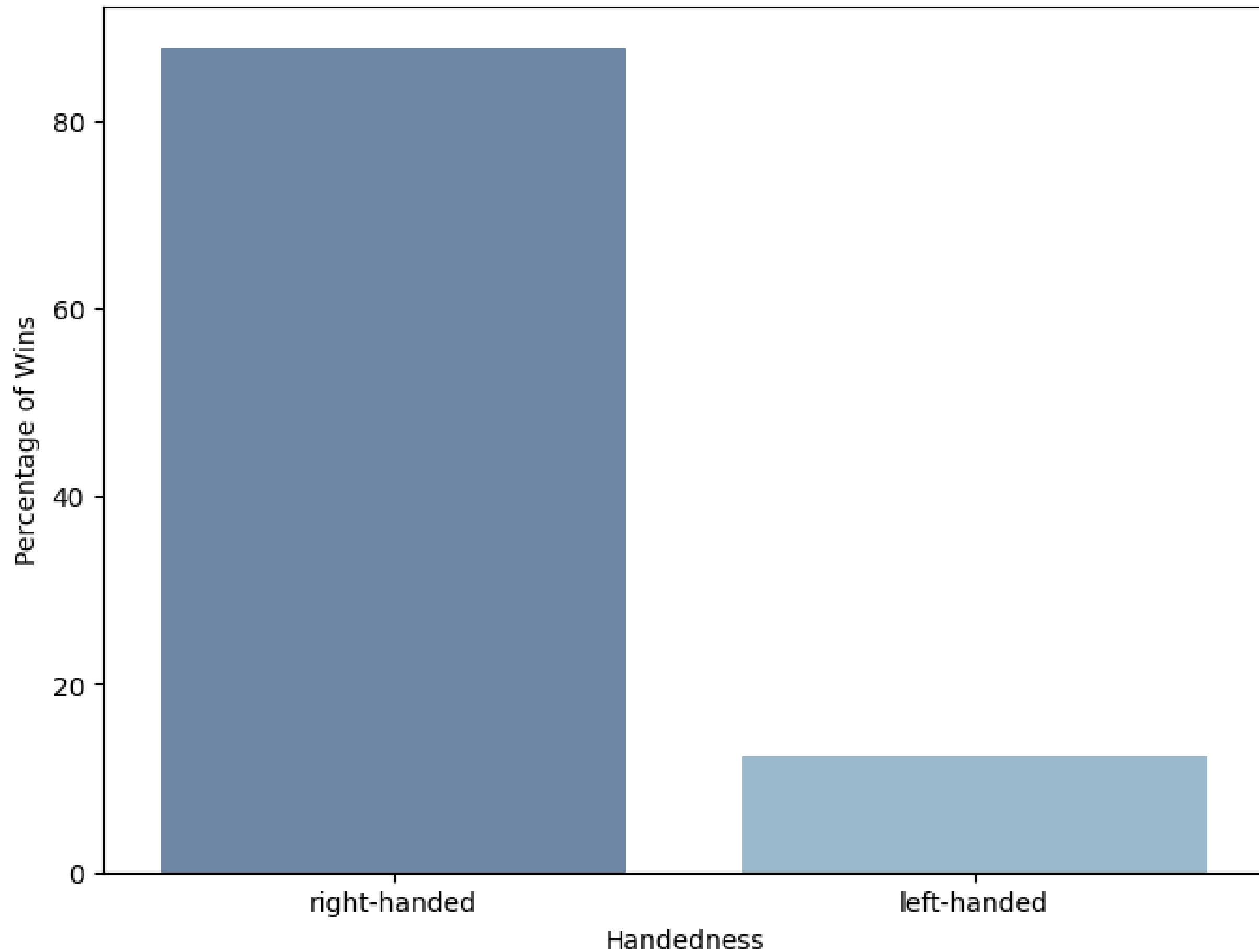


winner_plays

right-handed 87.744843%

left-handed 12.255157%

Winning Percentage by Player Handedness



Question 6 :

Which country has produced the most successful tennis players?

Top 10 countries by number of match wins:

France 992

Italy 930

USA 889

Russia 621

Germany 557

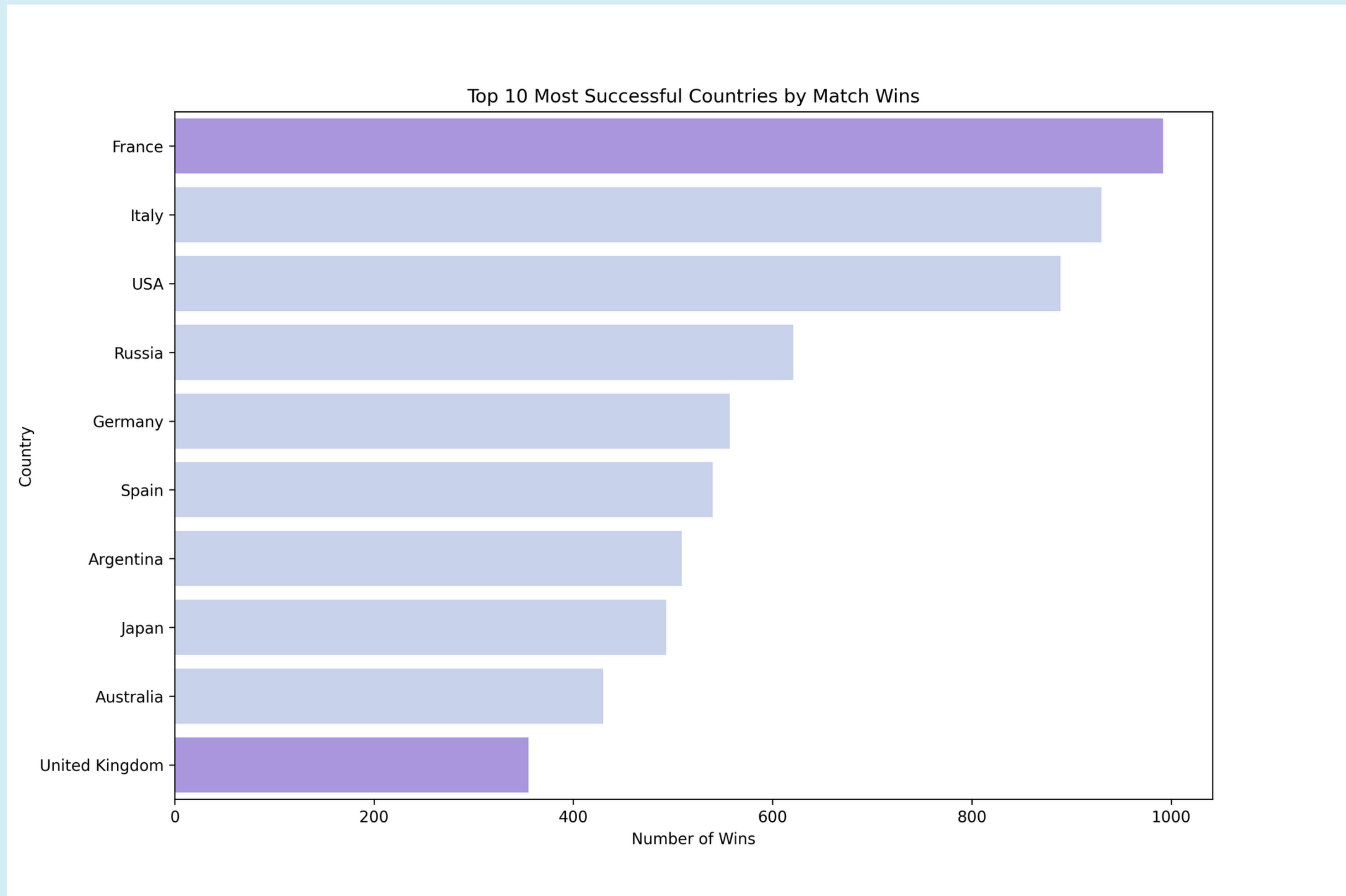
Spain 540

Argentina 509

Japan 493

Australia 430

United Kingdom 355



Top 10 countries with the best (lowest) current ranks: Poland 1.0

Serbia 1.0

Belarus 2.0

Italy 2.0

Spain 2.0

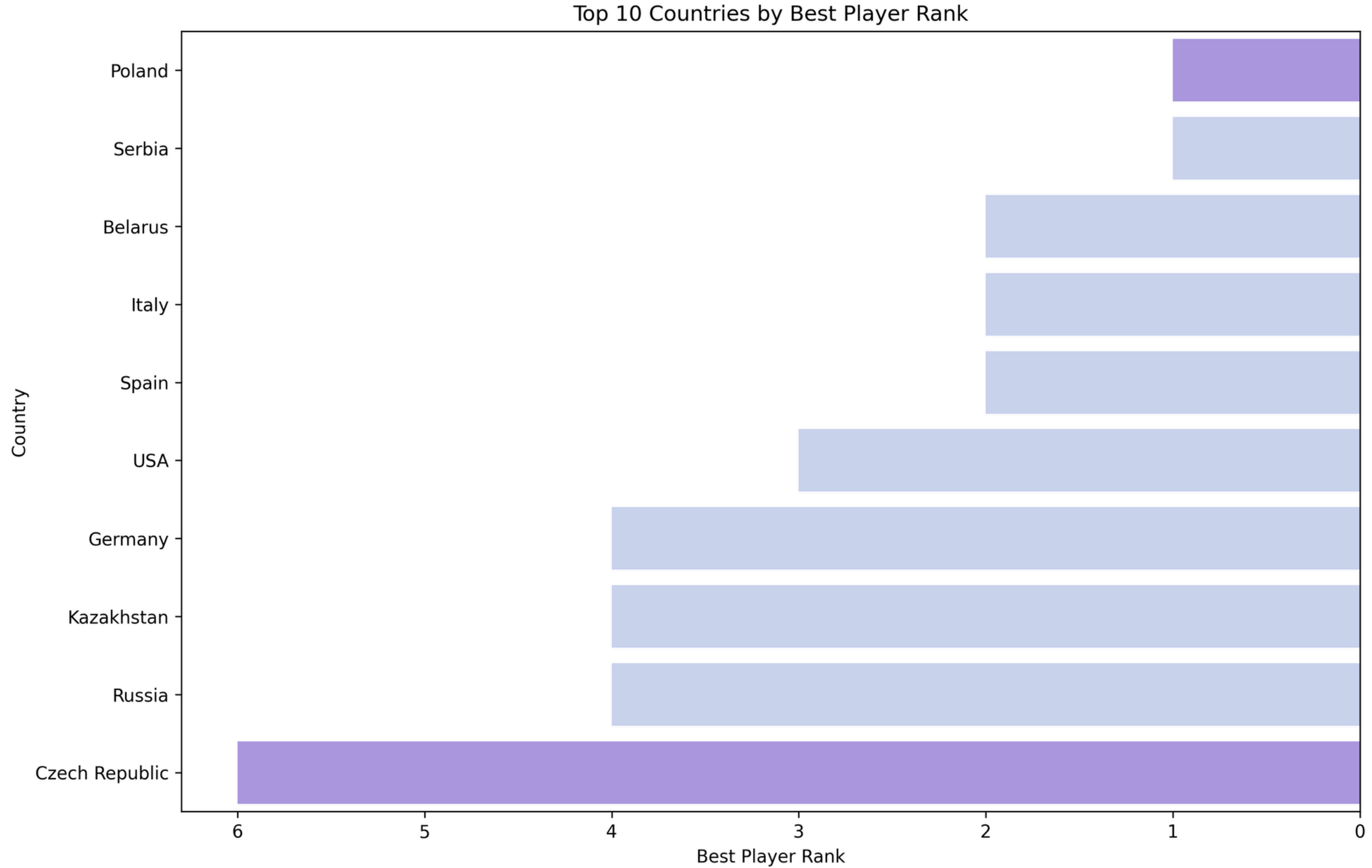
USA 3.0

Germany 4.0

Kazakhstan 4.0

Russia 4.0

Czech Republic 6.0

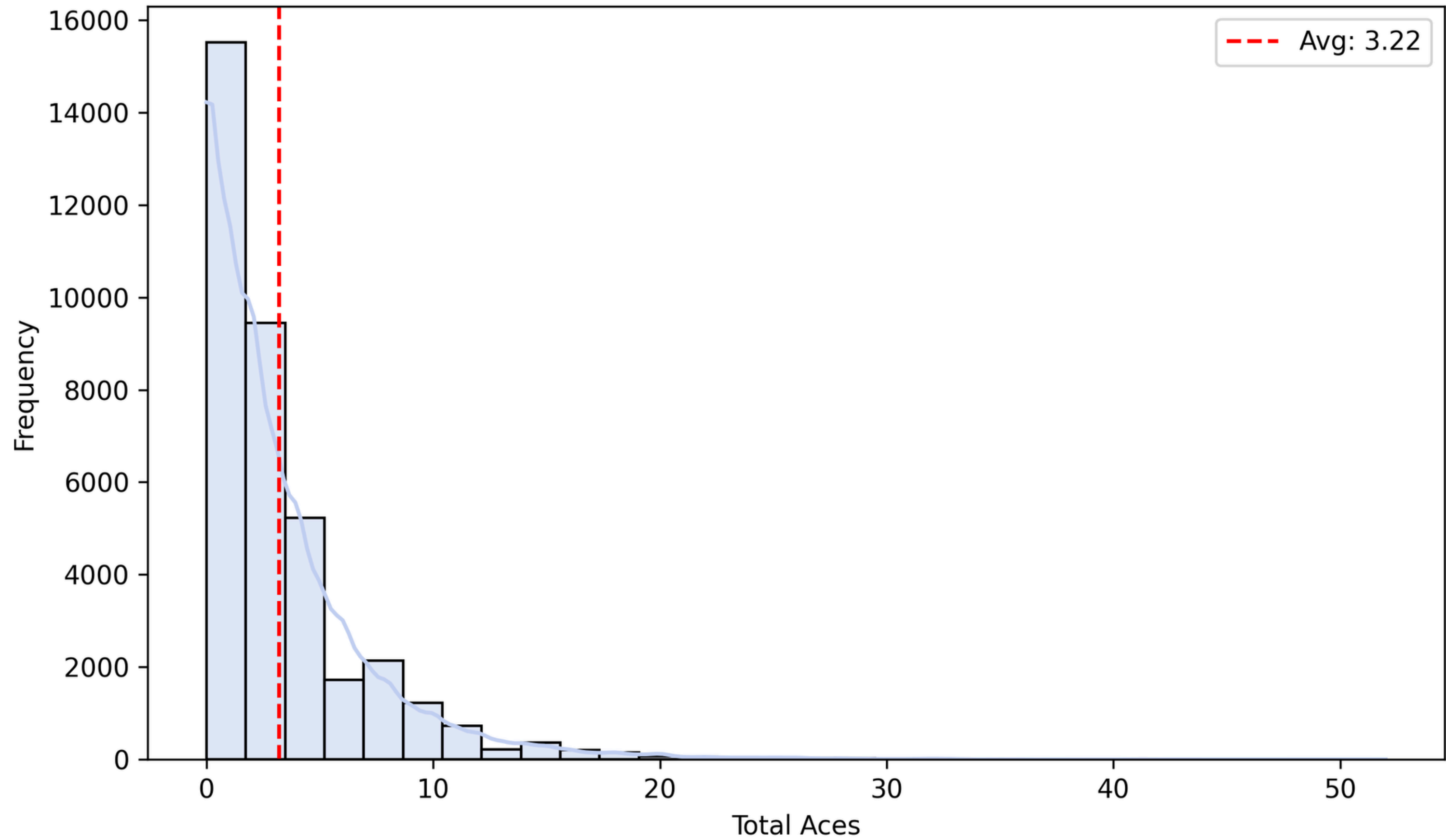


Question 7 :

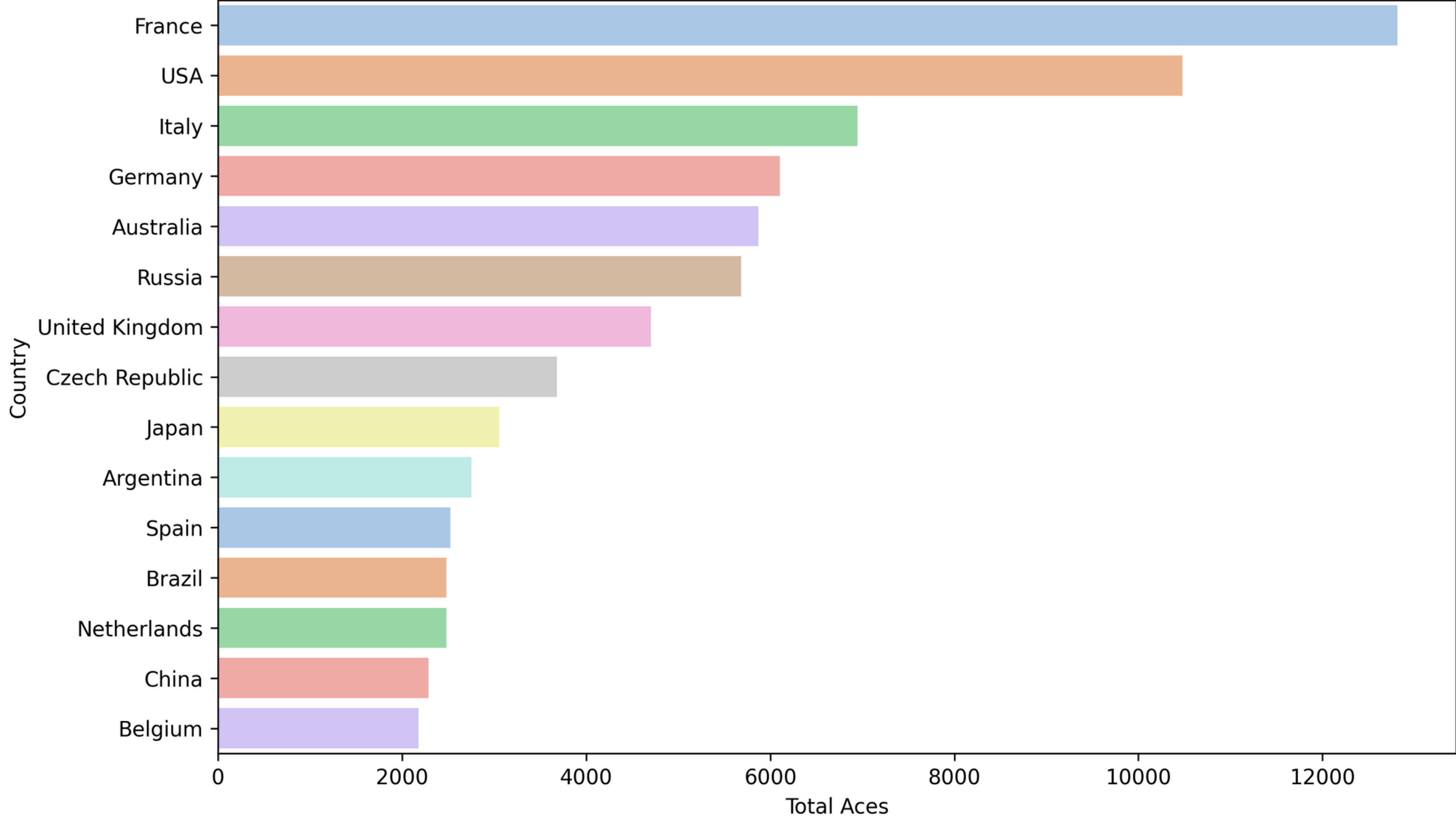
What is the average number of aces per match?

Average number of aces per match: 3.43

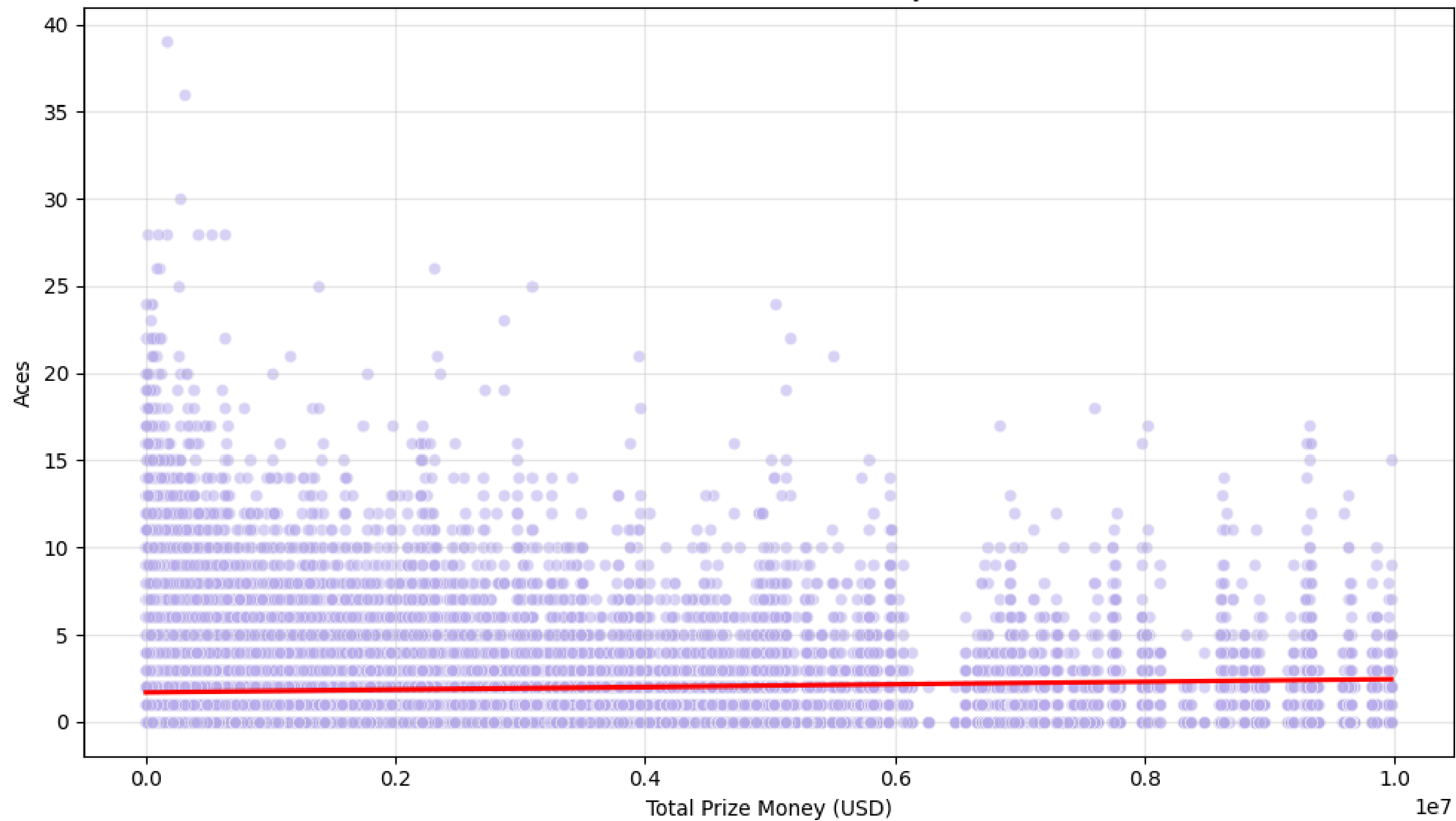
Distribution of Total Aces per Match



Top 15 Countries by Total Aces



Aces vs Total Prize Money

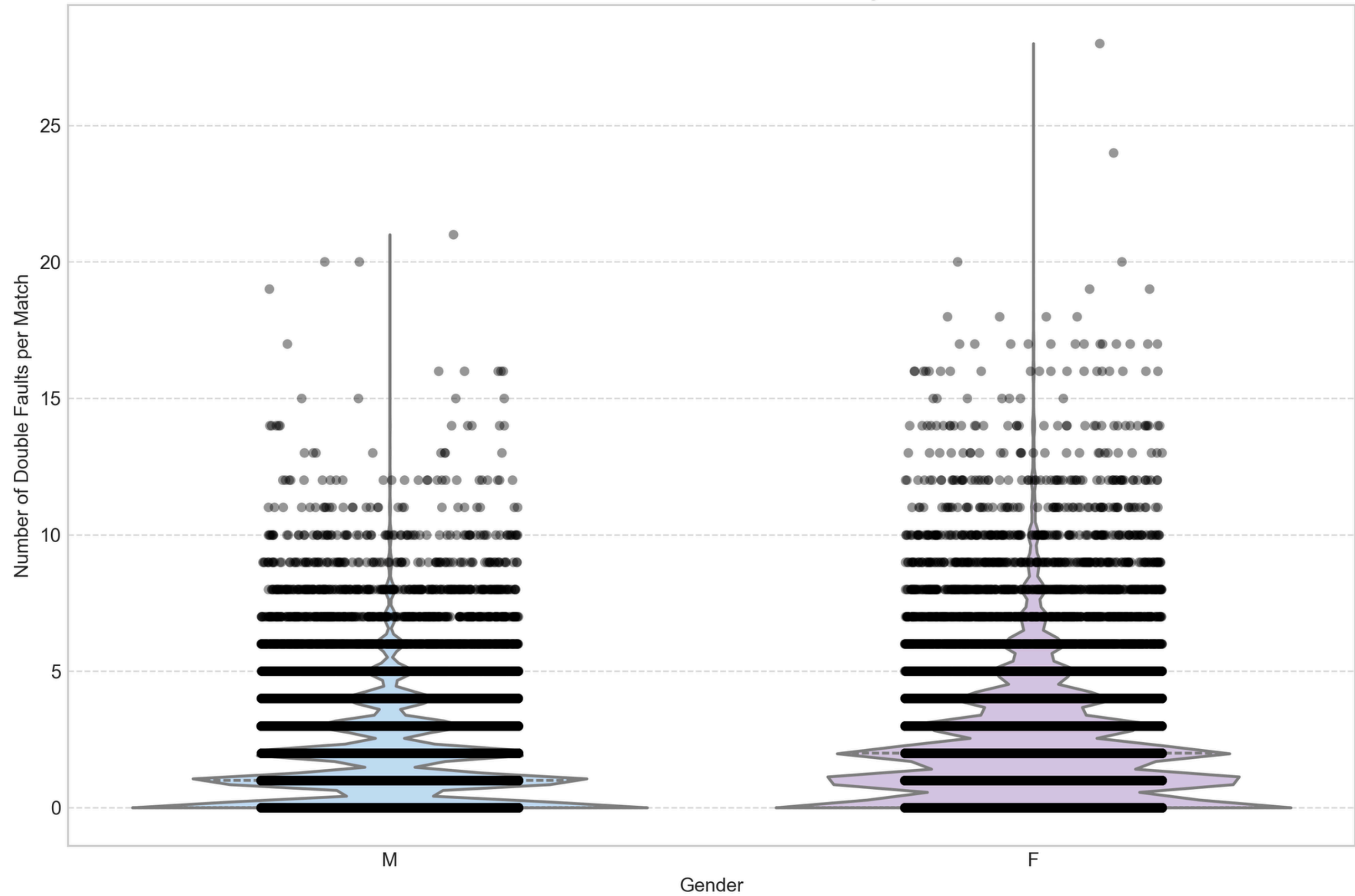


Question 8 :

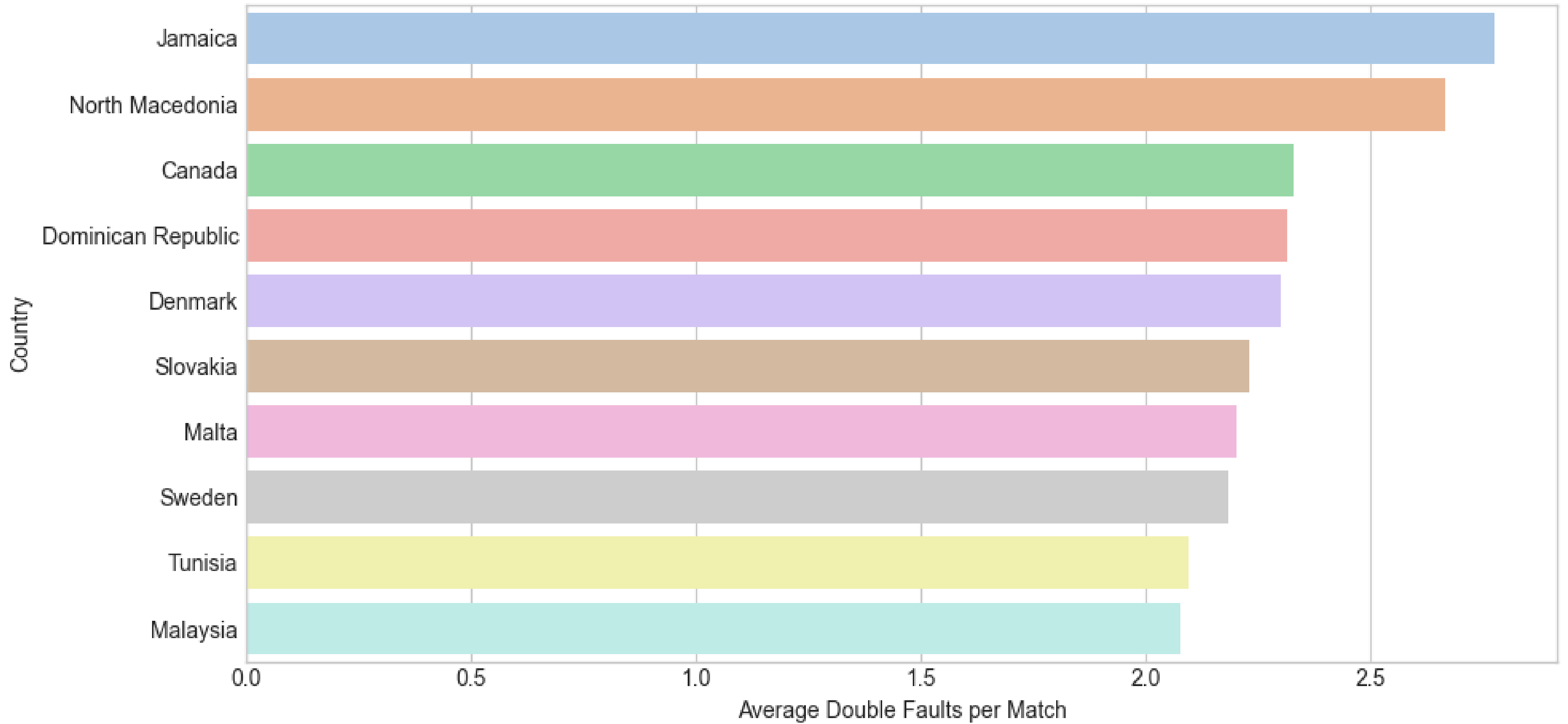
Is there a difference in the number of double faults based on gender?

Descriptive Statistics by Gender								
	count	mean	std	min	25%	50%	75%	max
F	30950.0	2.107981	2.249022	0.0	0.0	2.0	3.0	28.0
M	40166.0	1.599064	1.801070	0.0	0.0	1.0	2.0	21.0

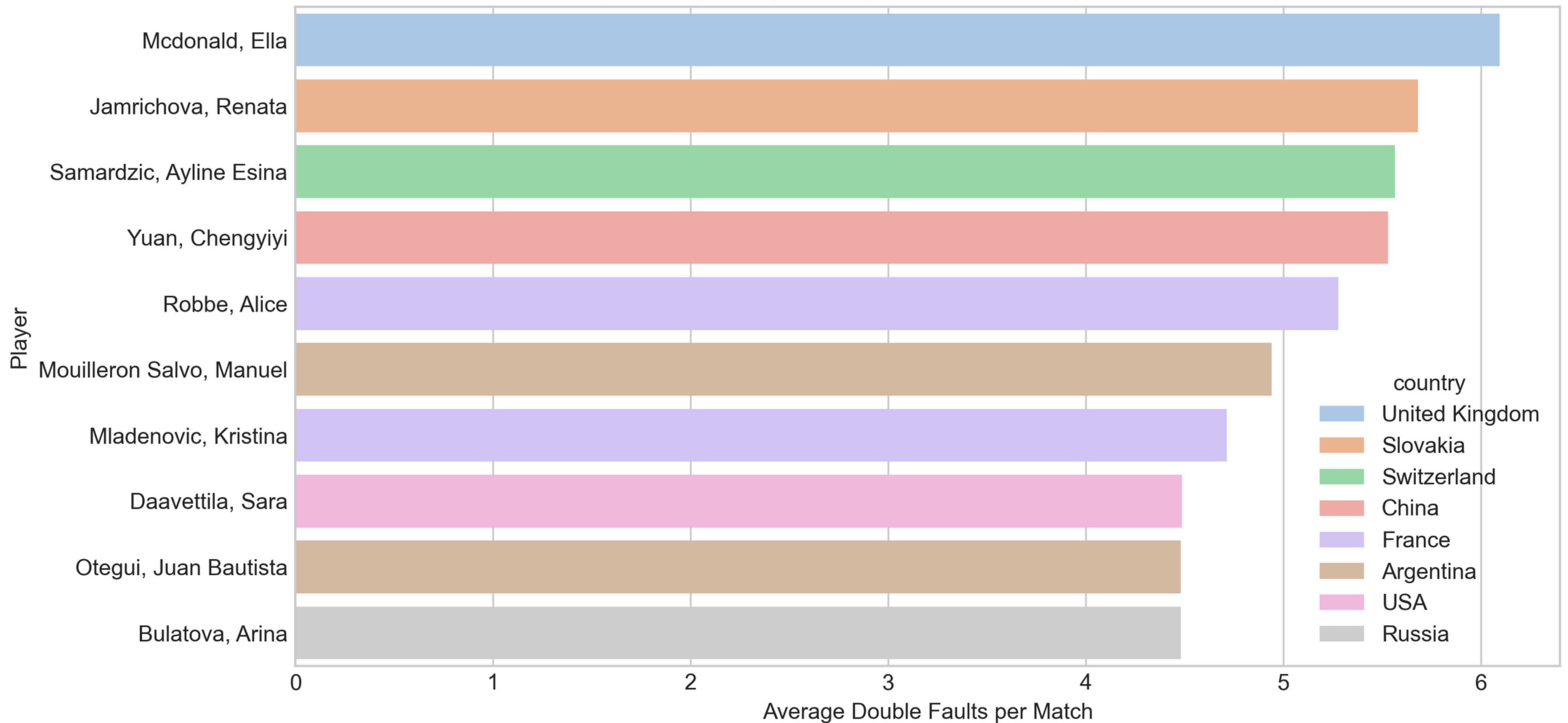
Distribution of Double Faults by Gender



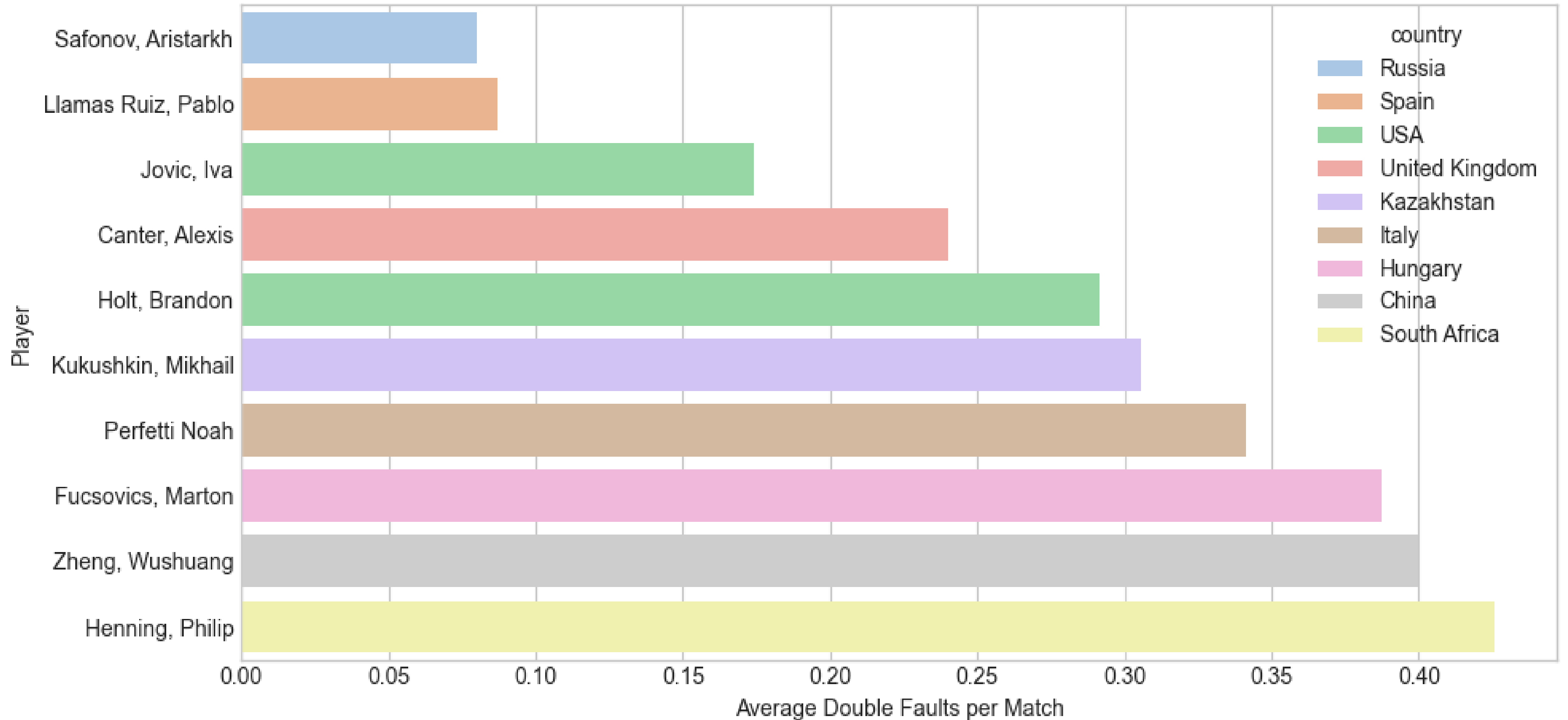
Top 10 Countries by Average Double Faults per Match



Top 10 Players by Average Double Faults (Min. 20 Matches)



Bottom 10 Players by Average Double Faults (Min. 20 Matches)

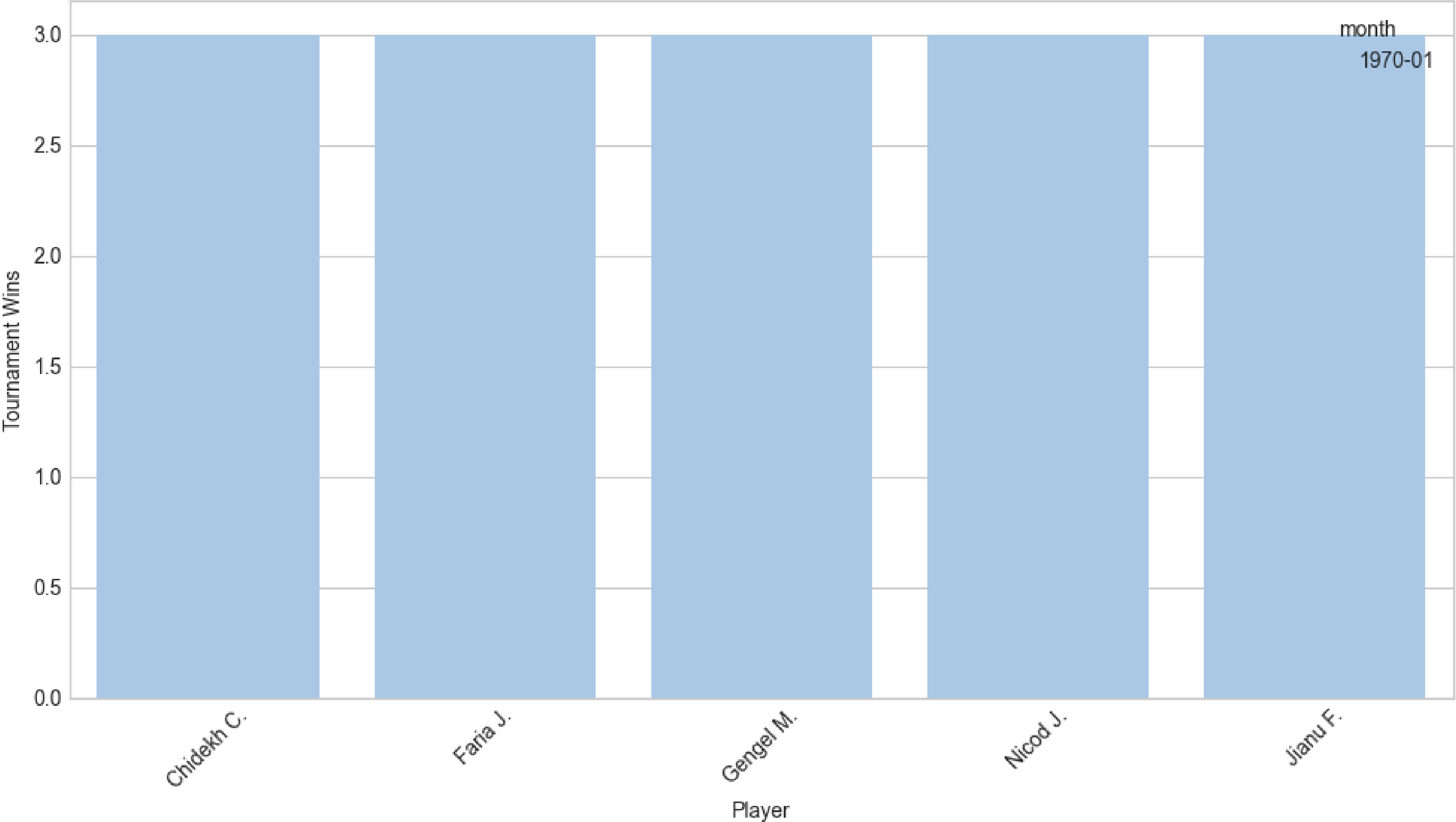


Question 9 :

Which player has won the most tournaments in a single month?

Player with most tournament wins in a single month:
winner Chidekh C.
month 1970-01
tournament_wins 3

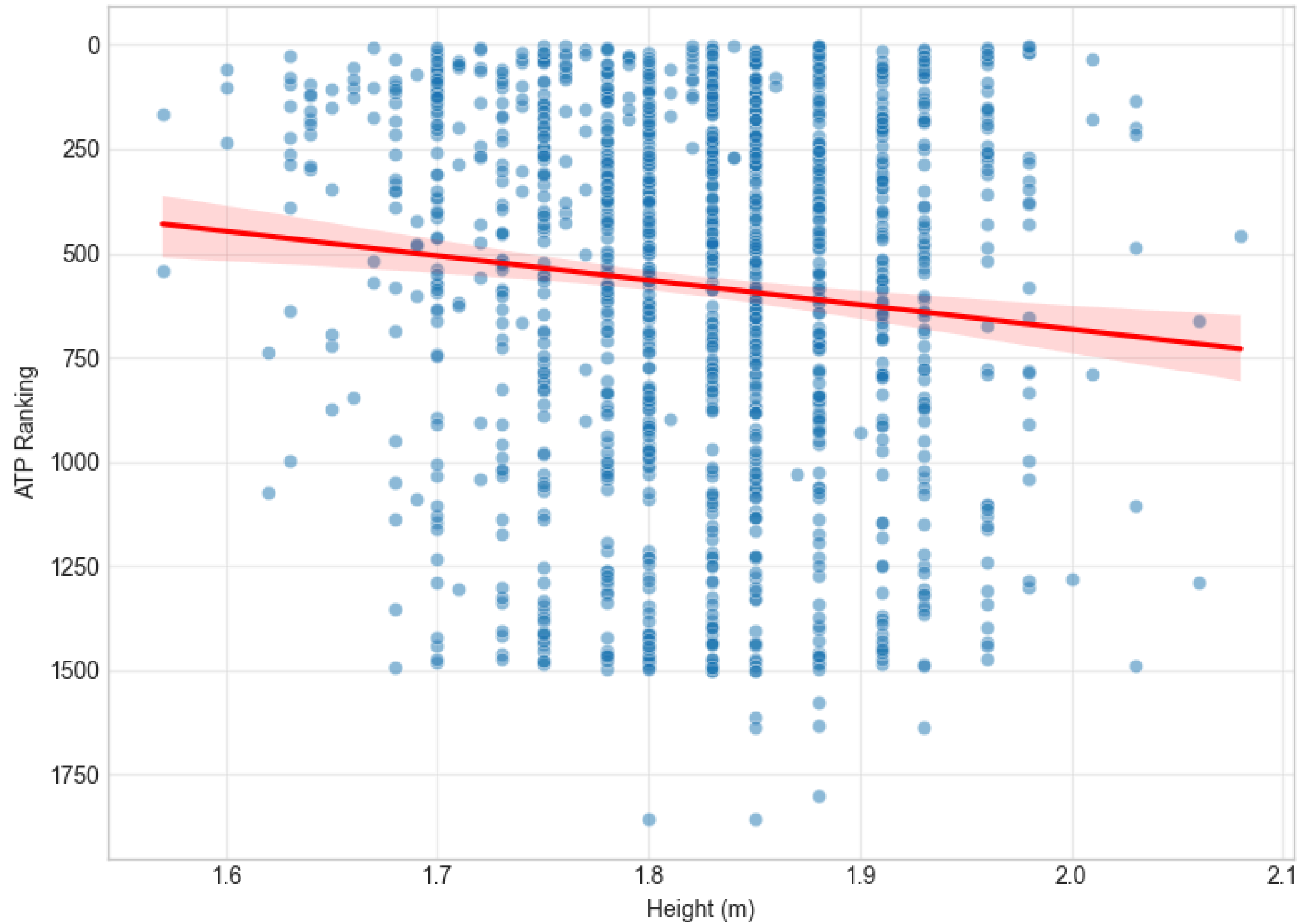
Top 5 Players by Tournament Wins in a Single Month



Question 10 :

Is there a correlation between a player's height and their ranking?

Height vs. Ranking



Question 11 :

What is the average duration of matches?

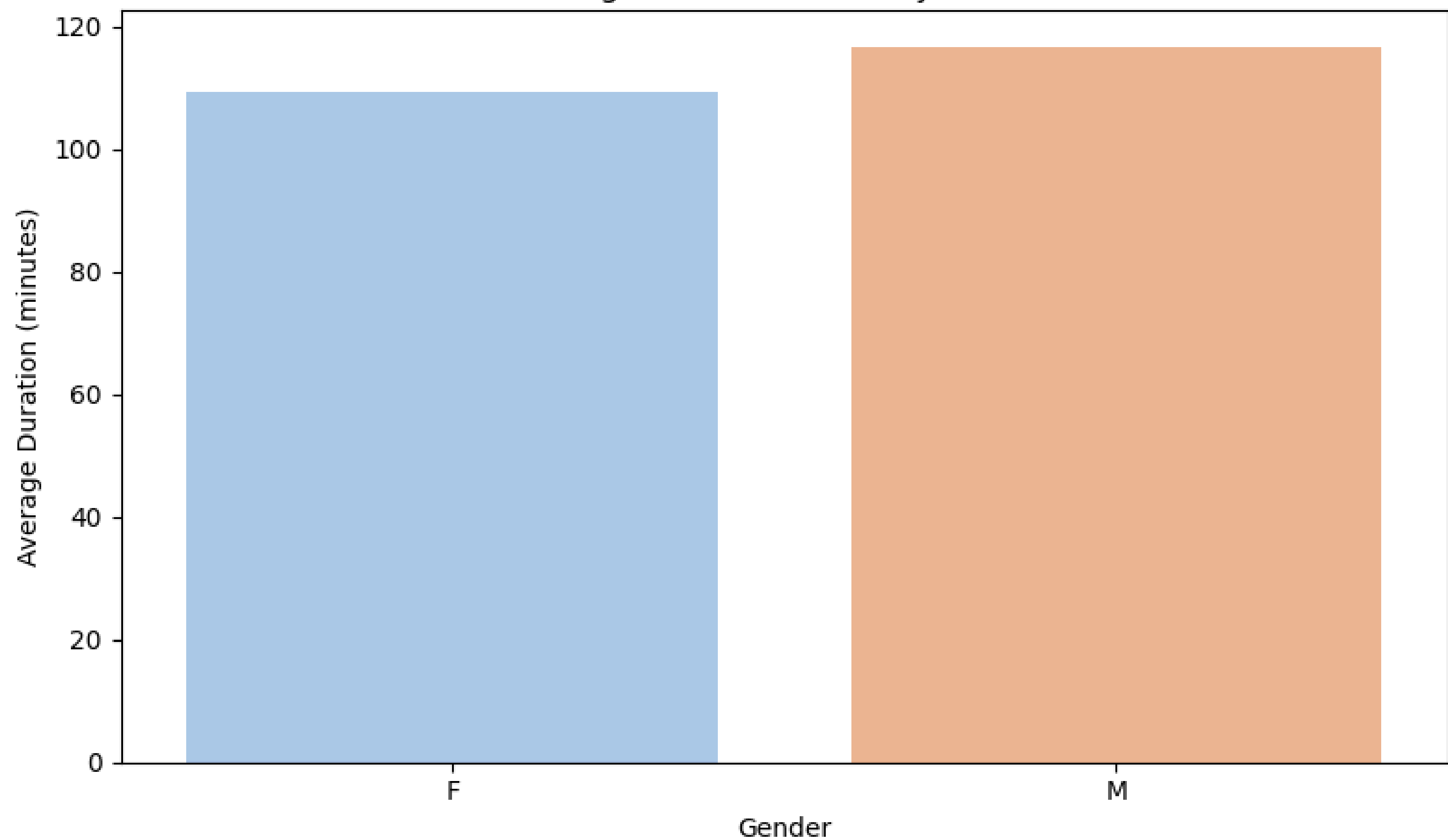
Average match duration (in minutes) by gender:

gender	total_duration_min
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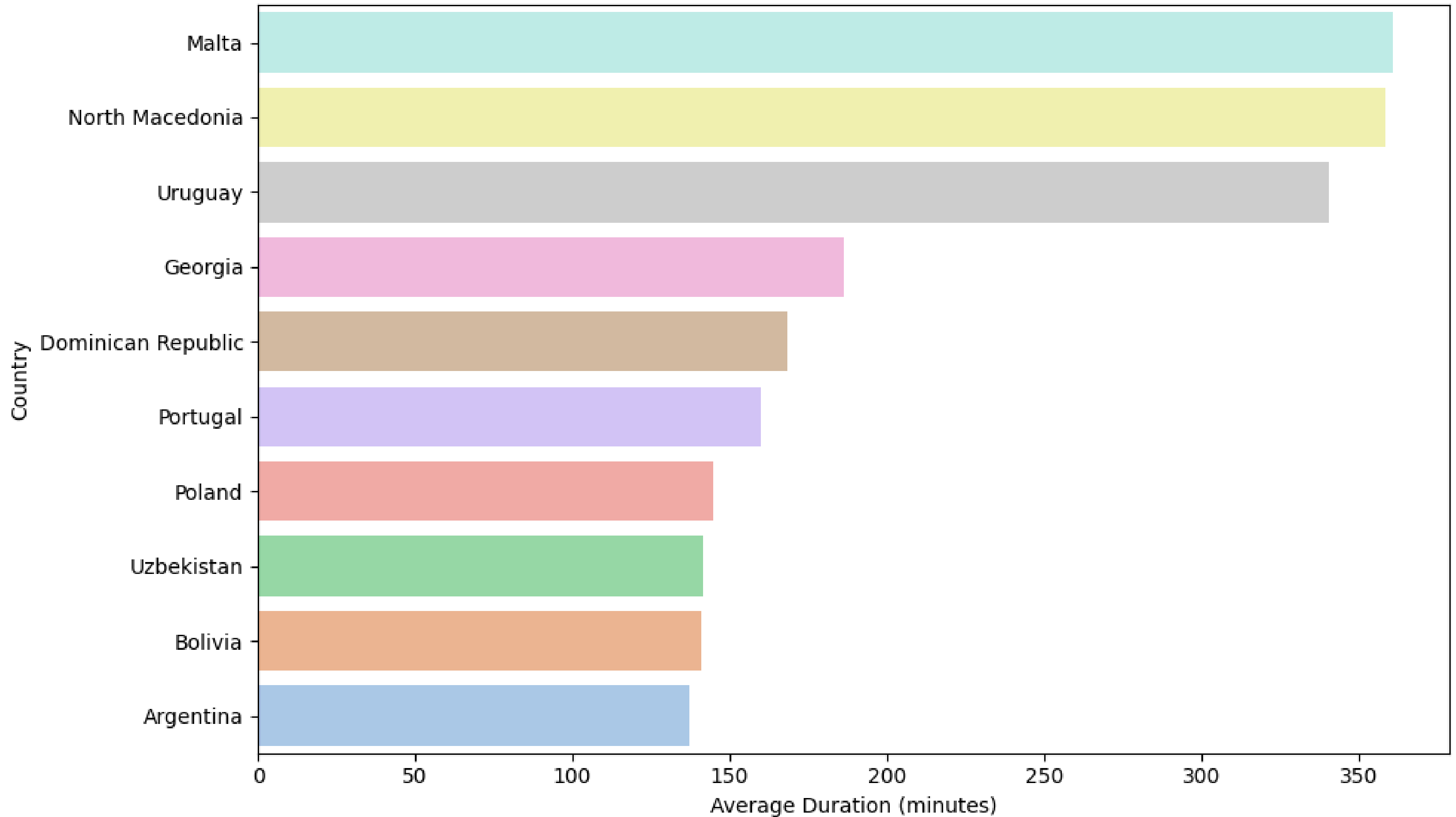
F	109.349740
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M	116.743139
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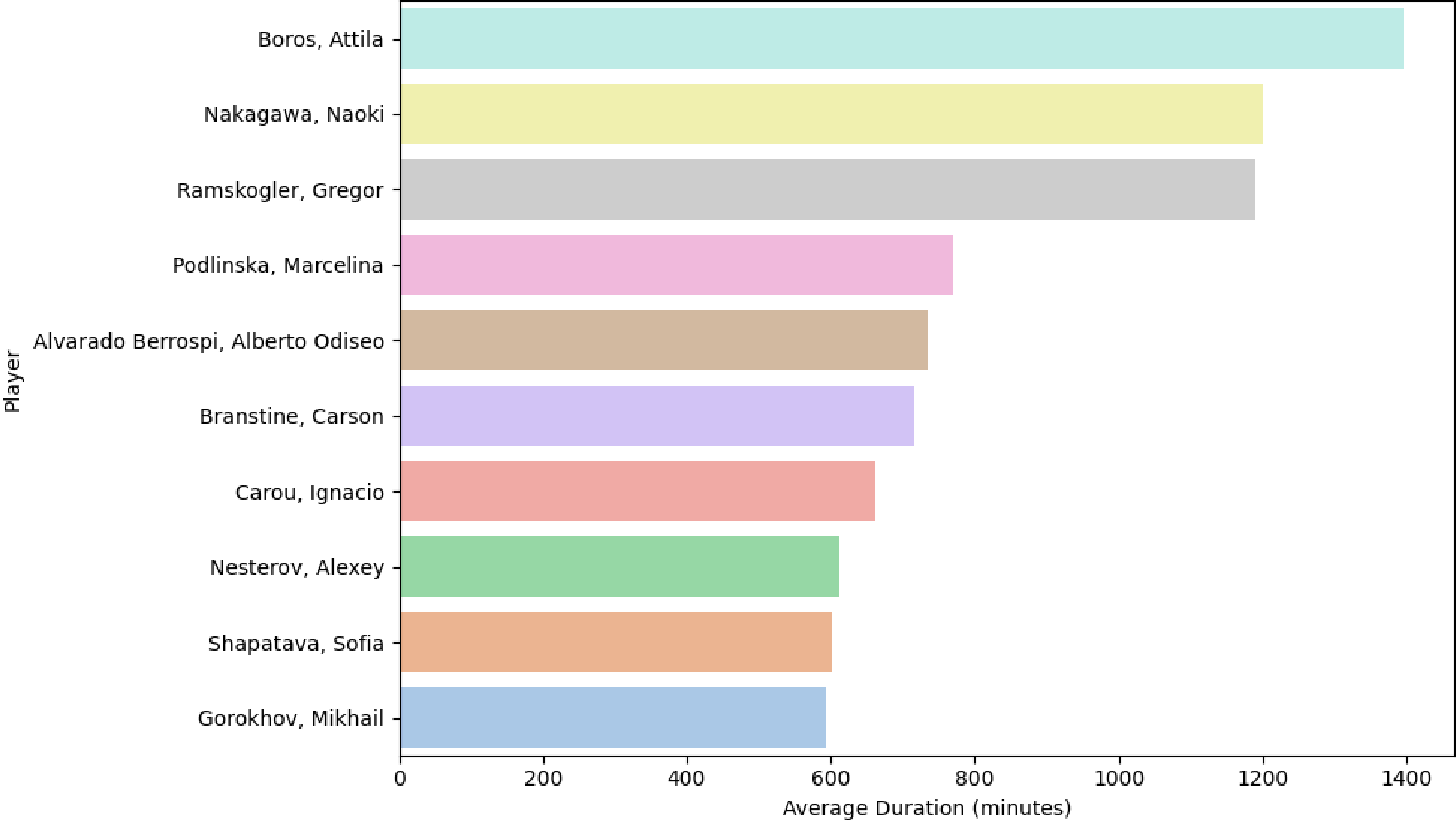
Average Match Duration by Gender



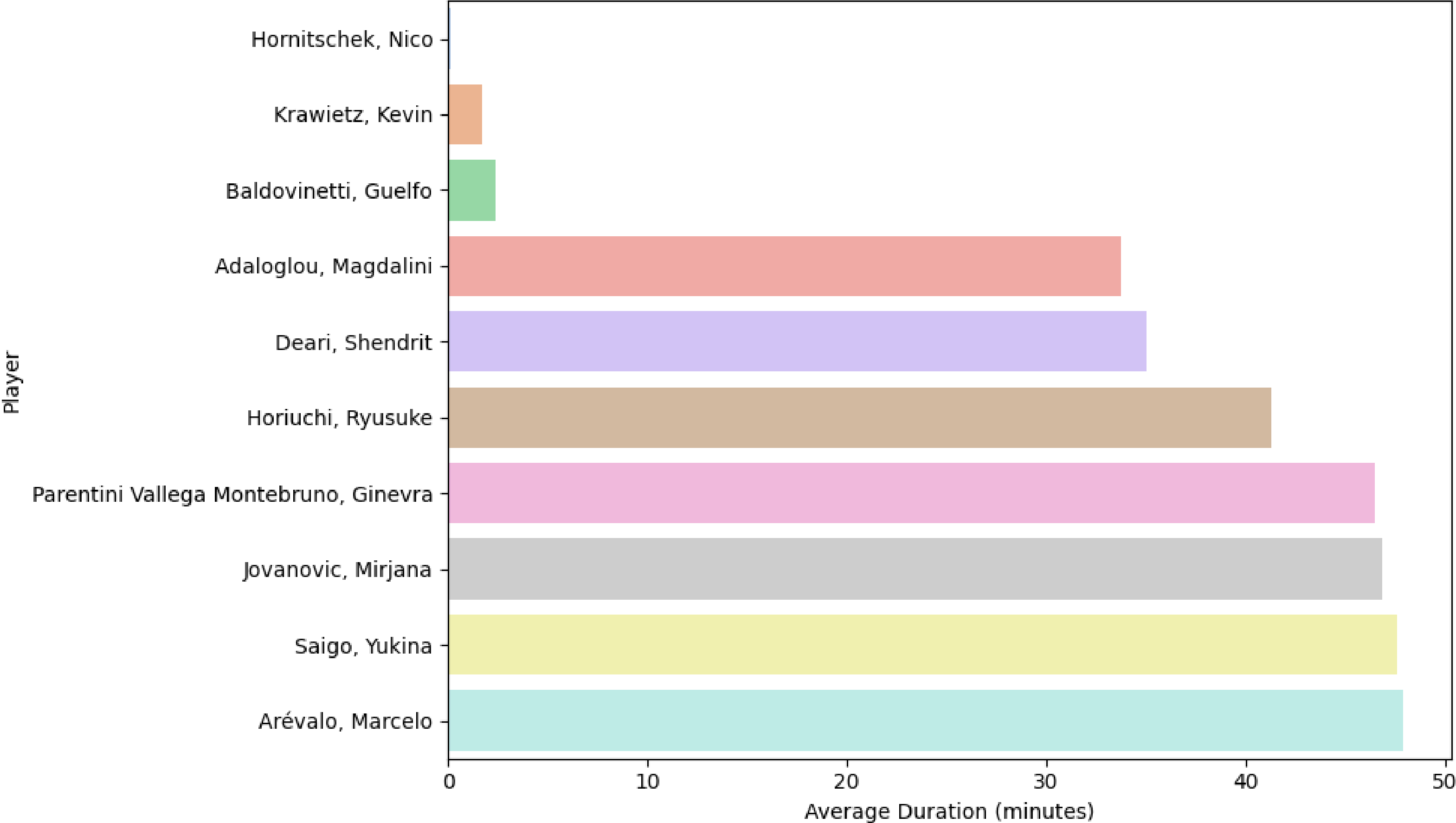
Top 10 Countries by Average Match Duration



Top 10 Players with Longest Average Match Duration



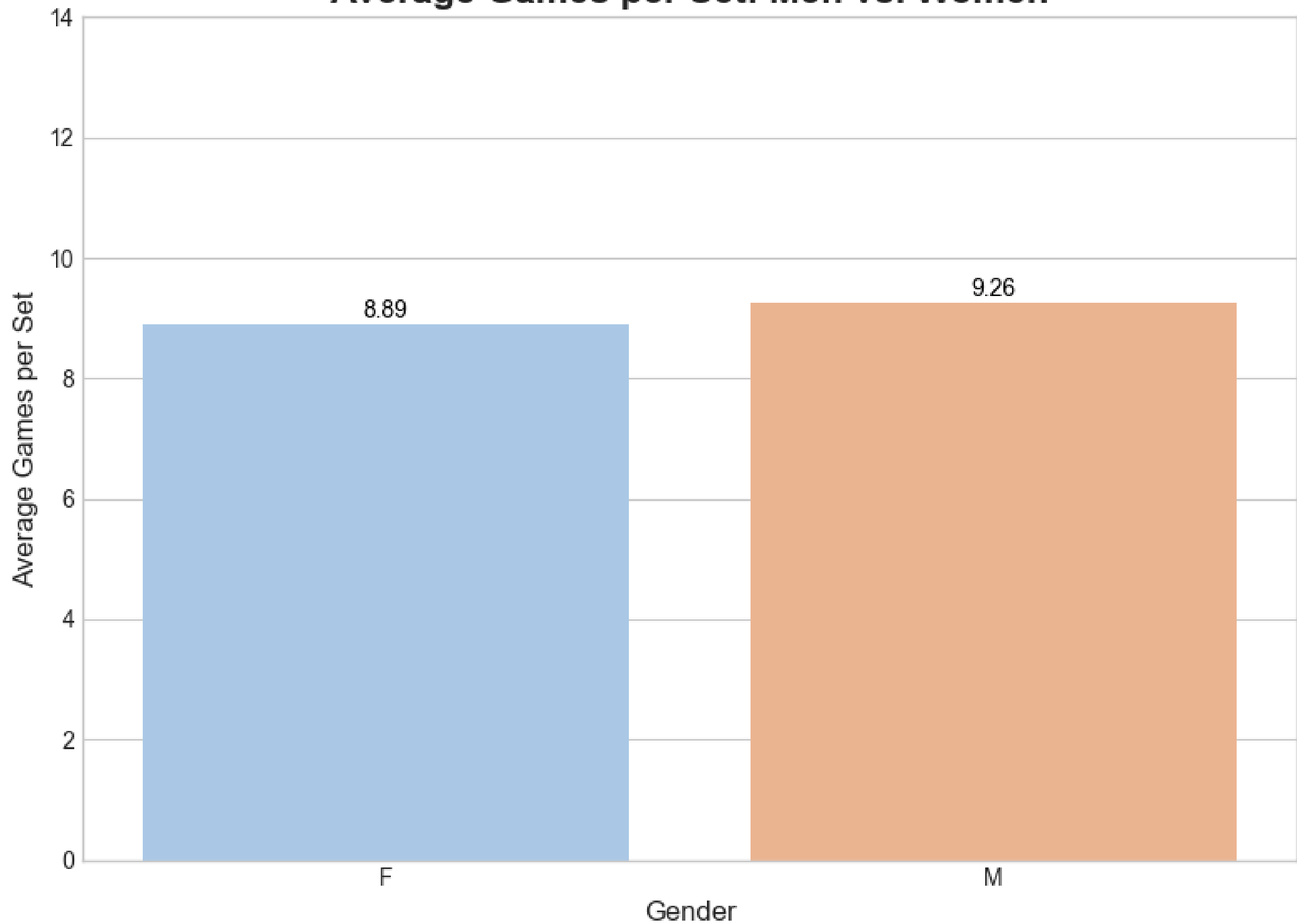
Top 10 Players with Shortest Average Match Duration



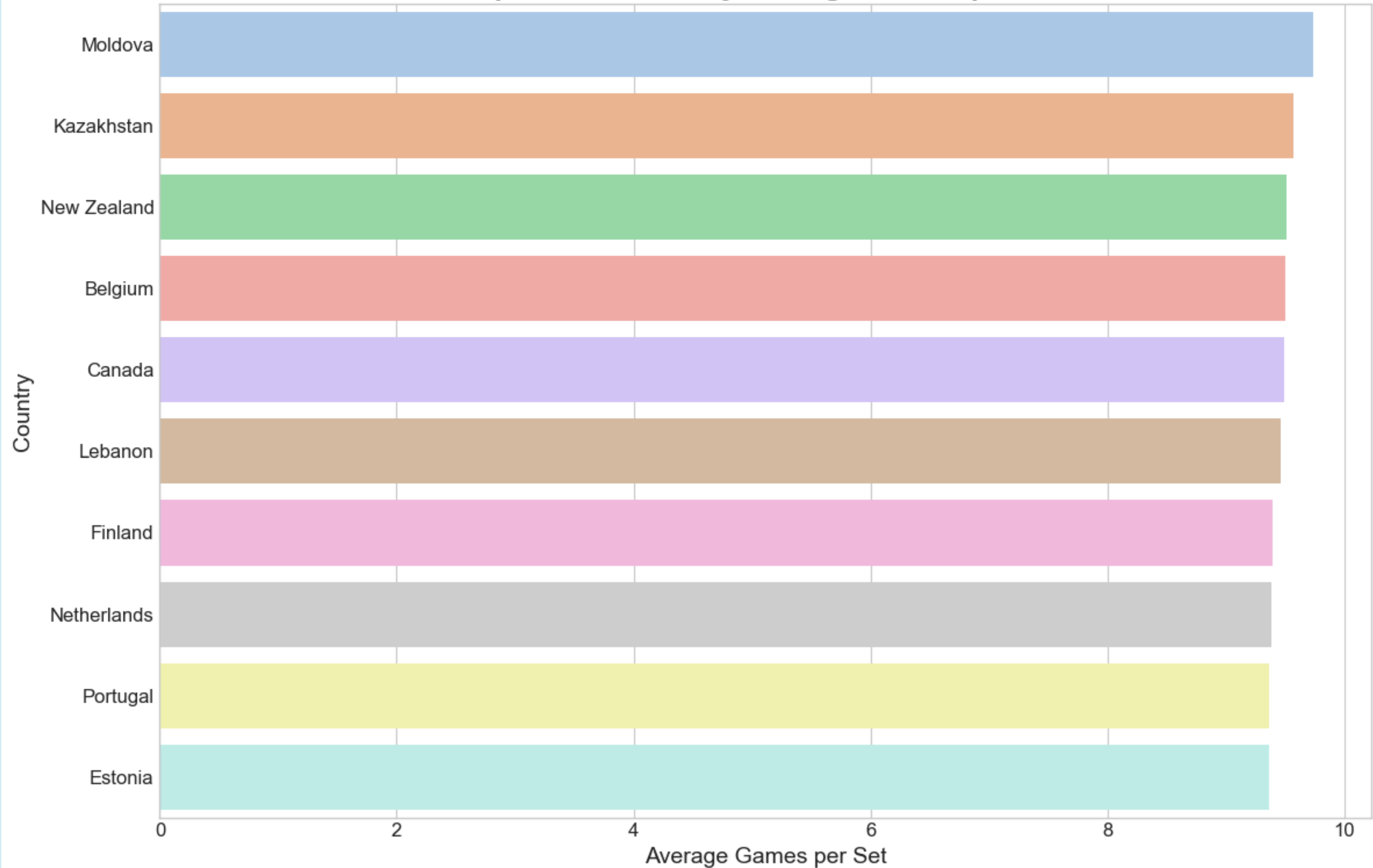
Question 12 :

What is the average number of games per set in men's matches compared to women's matches?

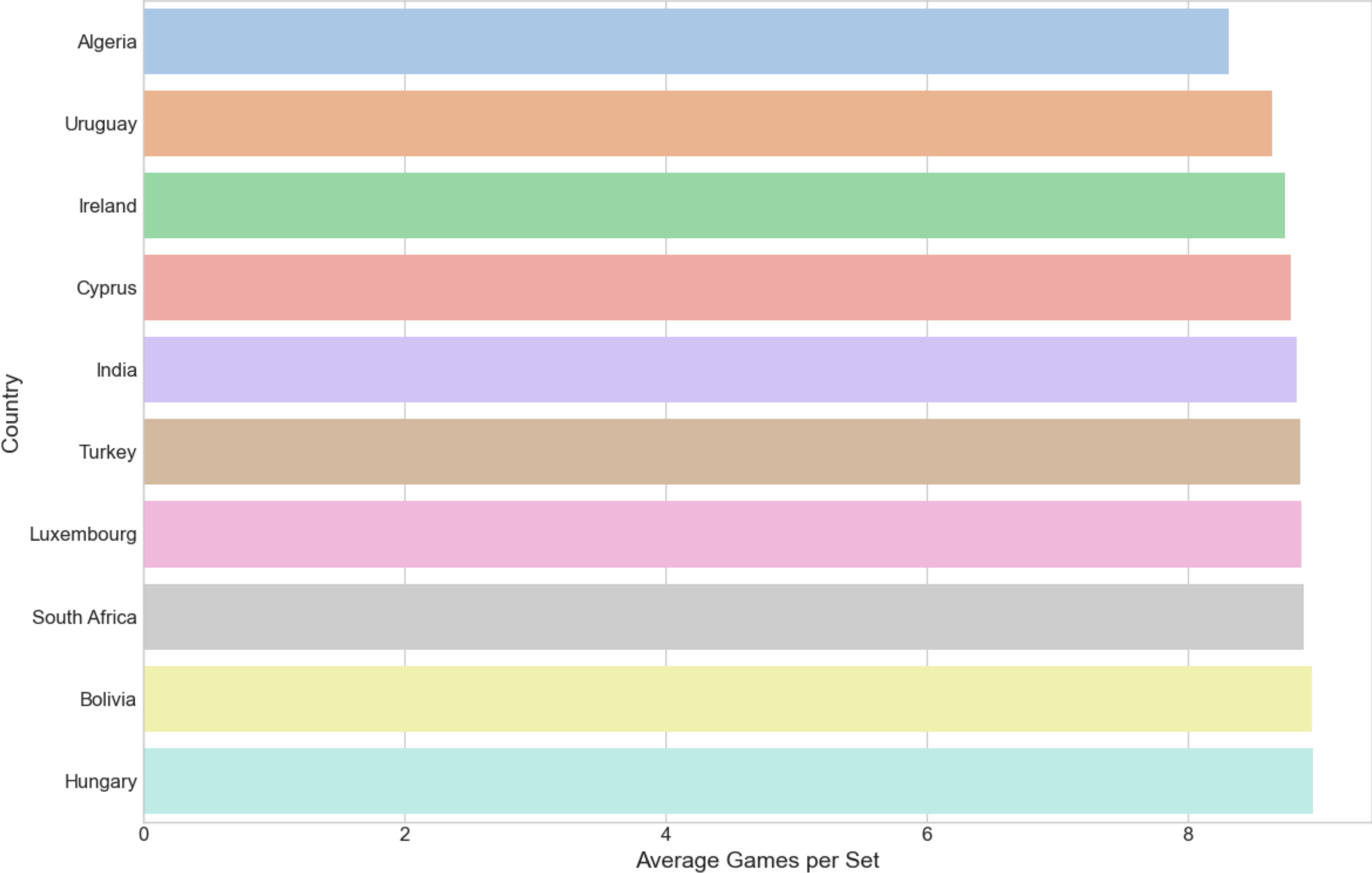
Average Games per Set: Men vs. Women



Top 10 Countries by Average Games per Set



Bottom 10 Countries by Average Games per Set



Question 13 :

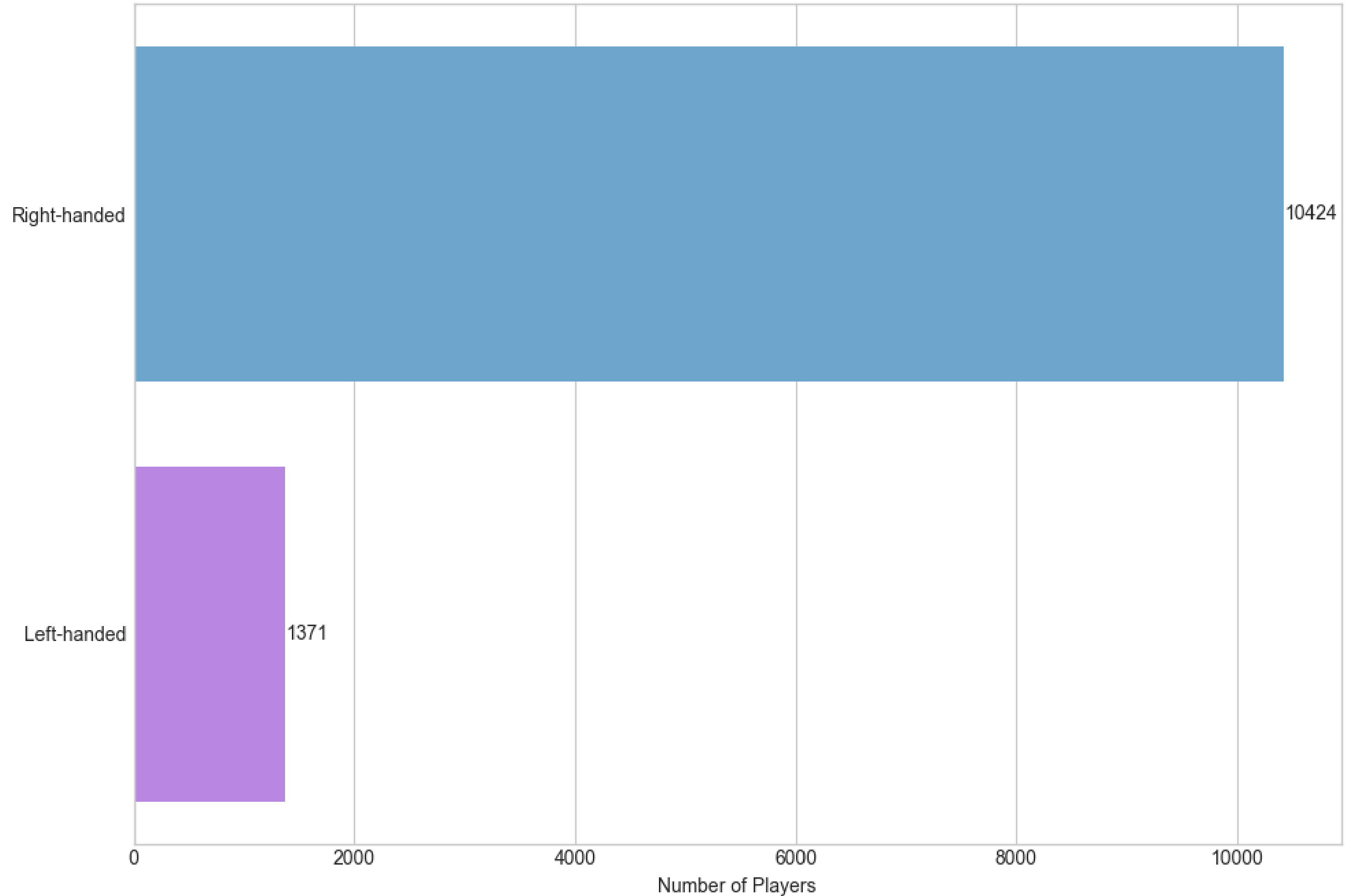
What is the distribution of left-handed versus right-handed players?

plays

Right-handed 10424

Left-handed 1371

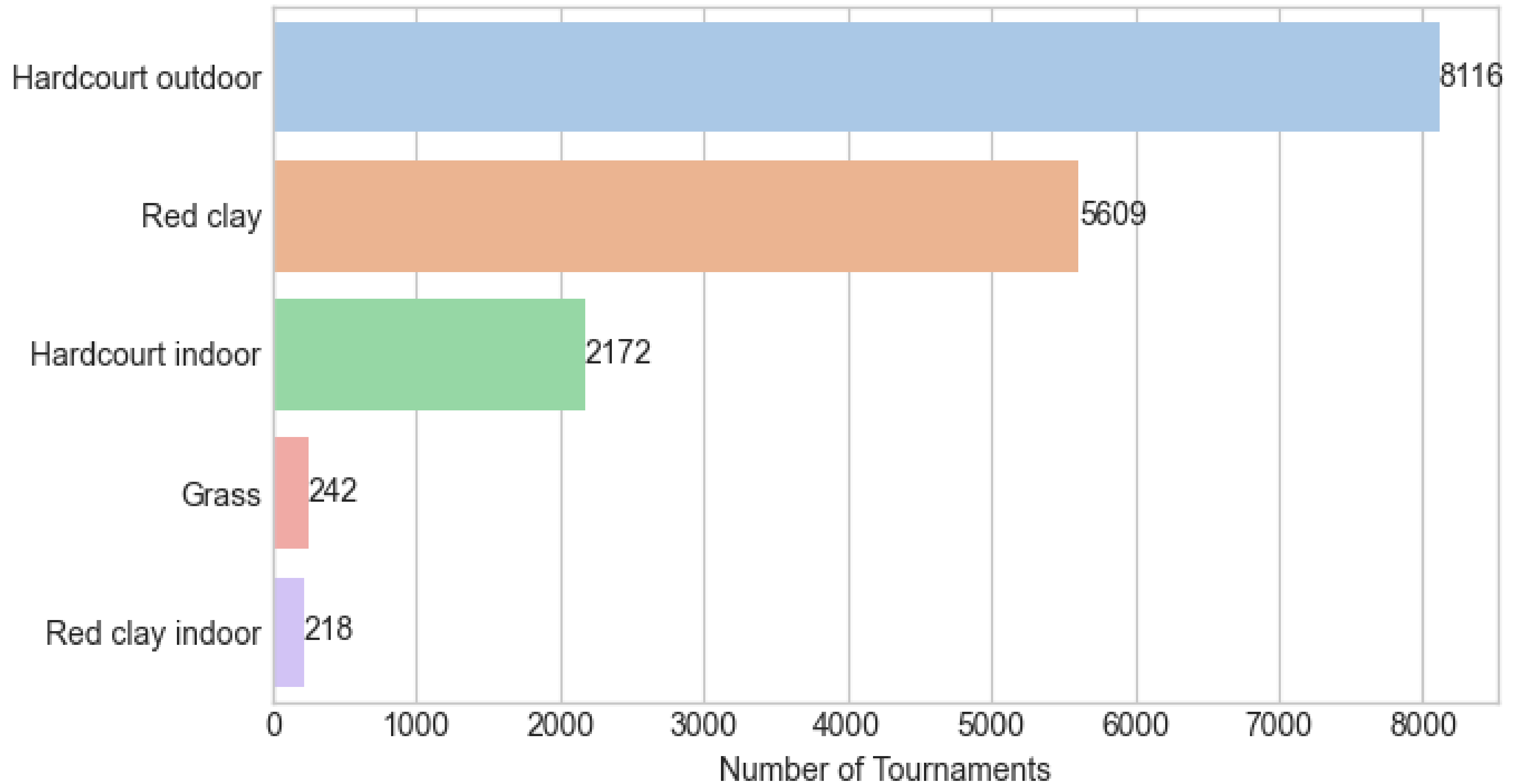
Left-handed vs Right-handed Players



Question 14 :

What is the most common type of surface used in tournaments?

Top 5 Tournament Surfaces

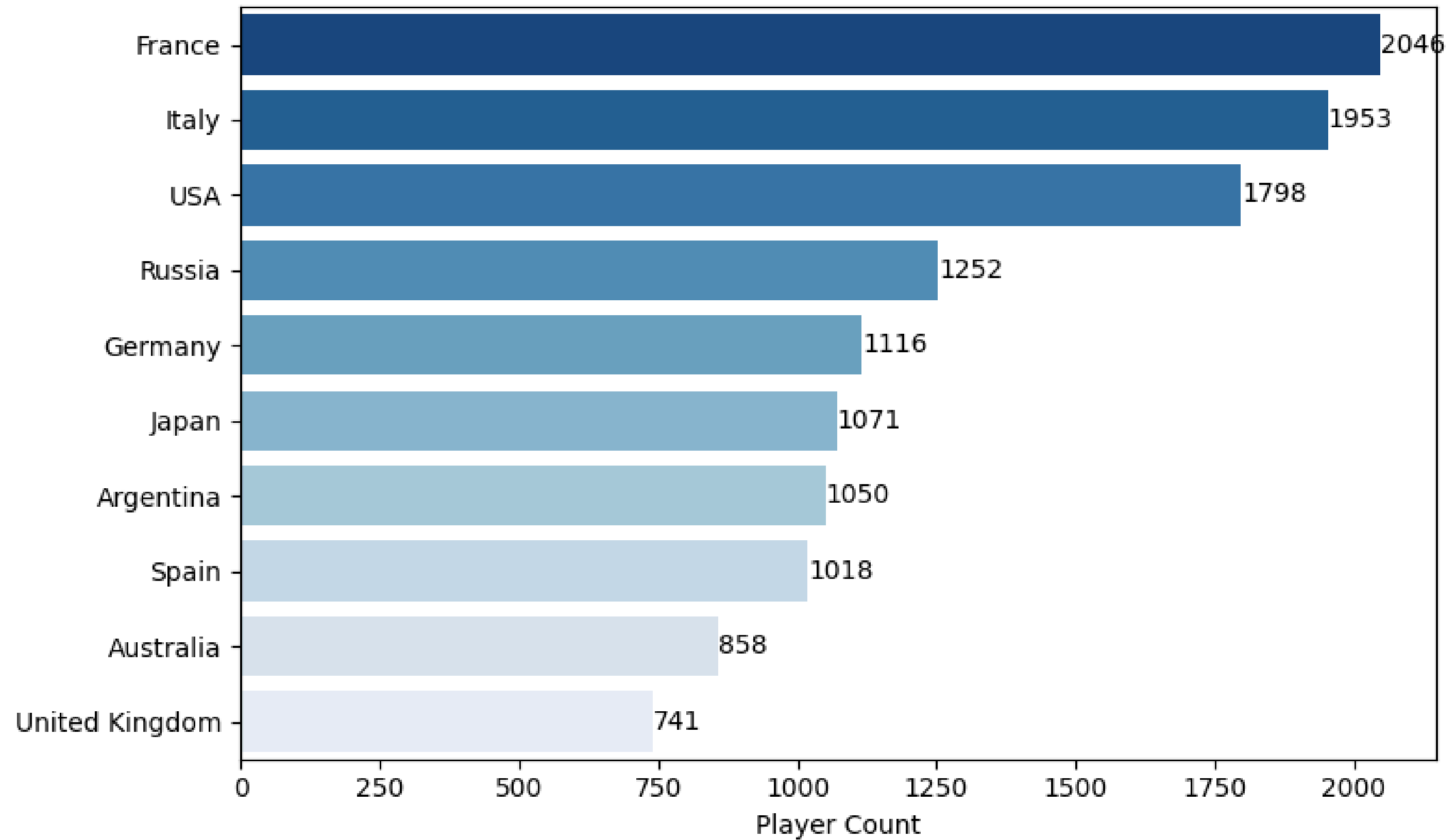


Question 15 :

How many distinct countries are represented in the dataset?

The number of distinct countries is: 101.

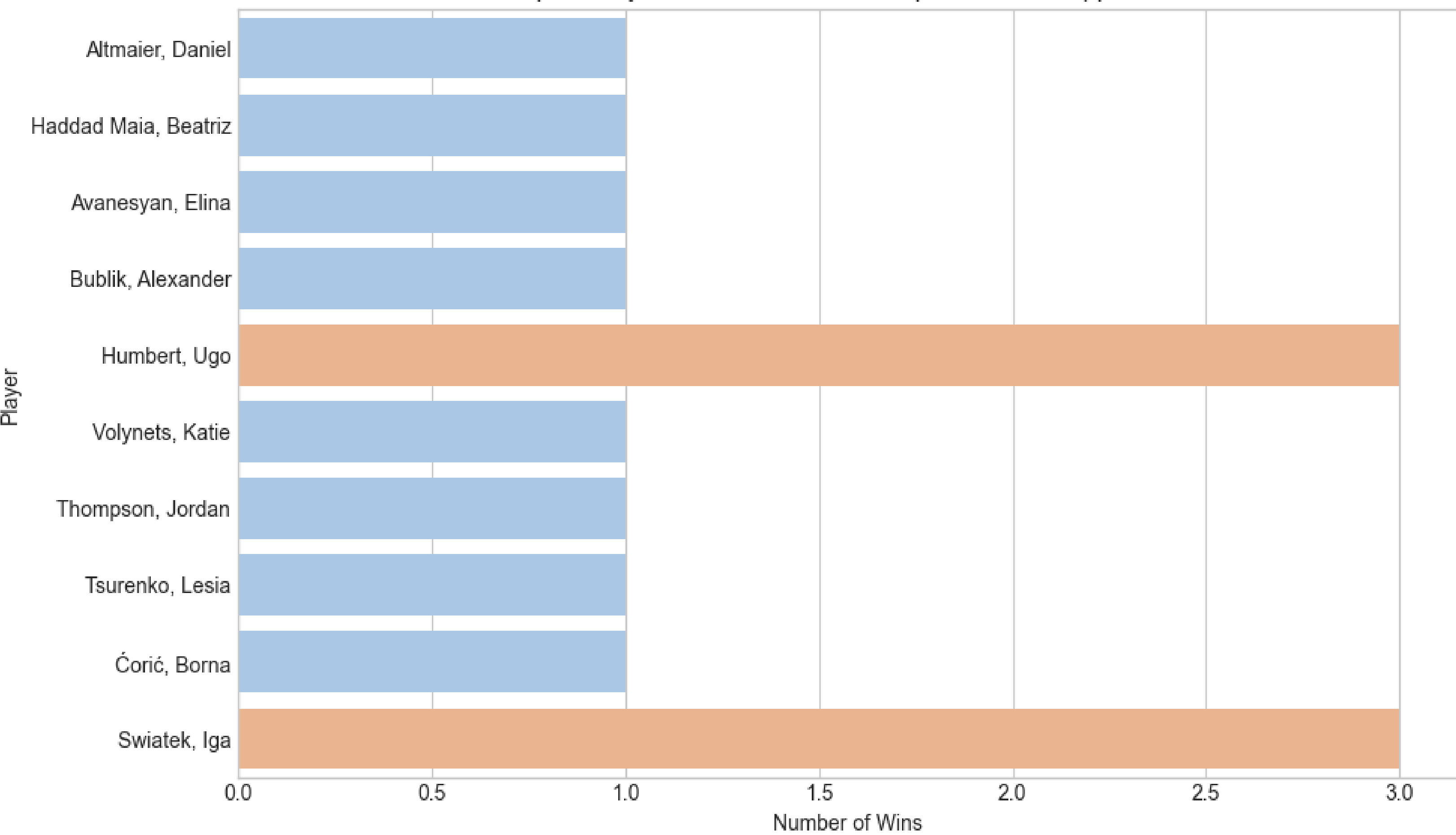
Top 10 Countries by Player Count



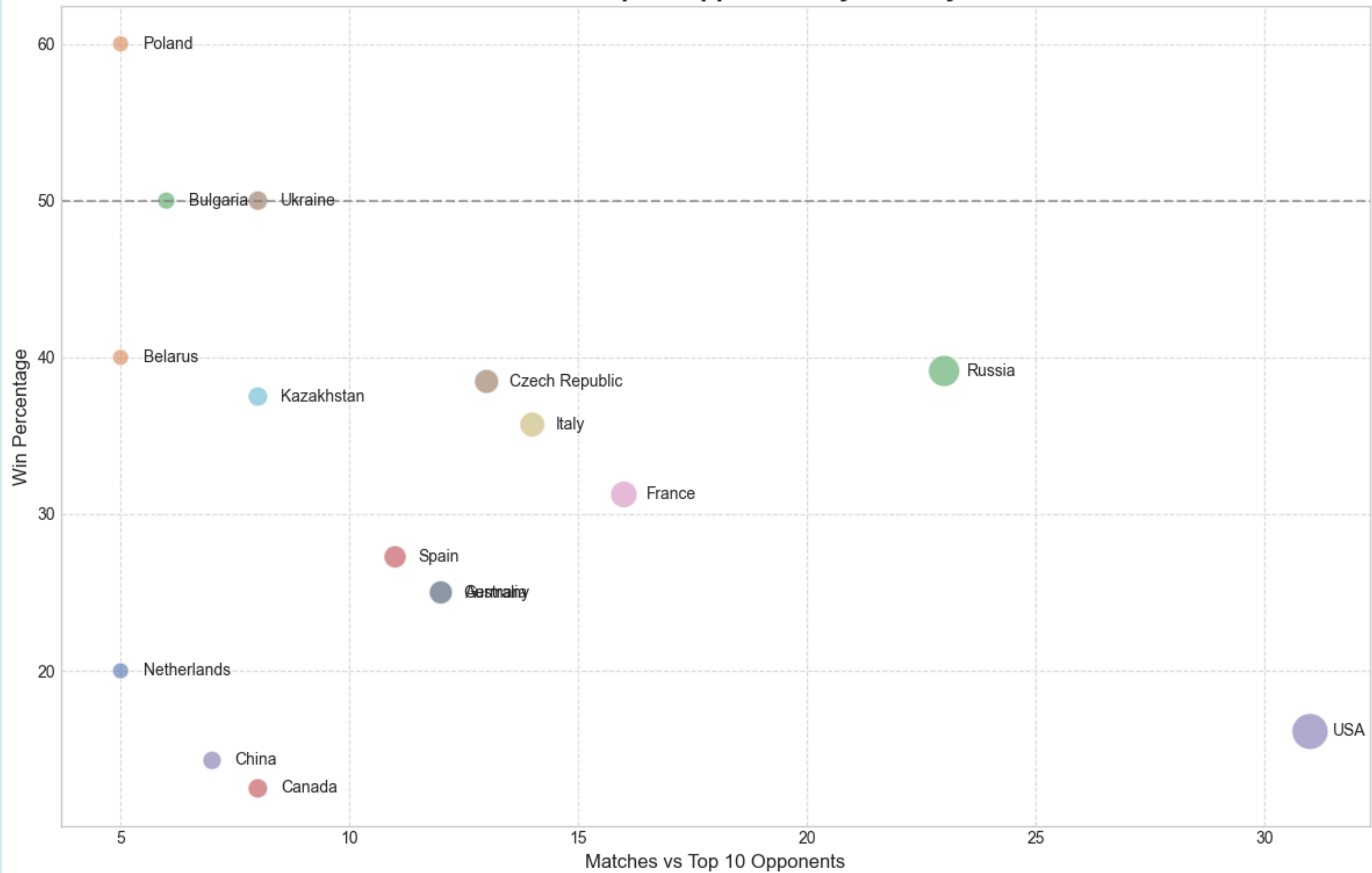
Question 16 :

Which player has the highest winning percentage against top 10 ranked opponents?

Top 10 Players with Most Wins vs Top 10 Ranked Opponents



Win % vs Top 10 Opponents by Country

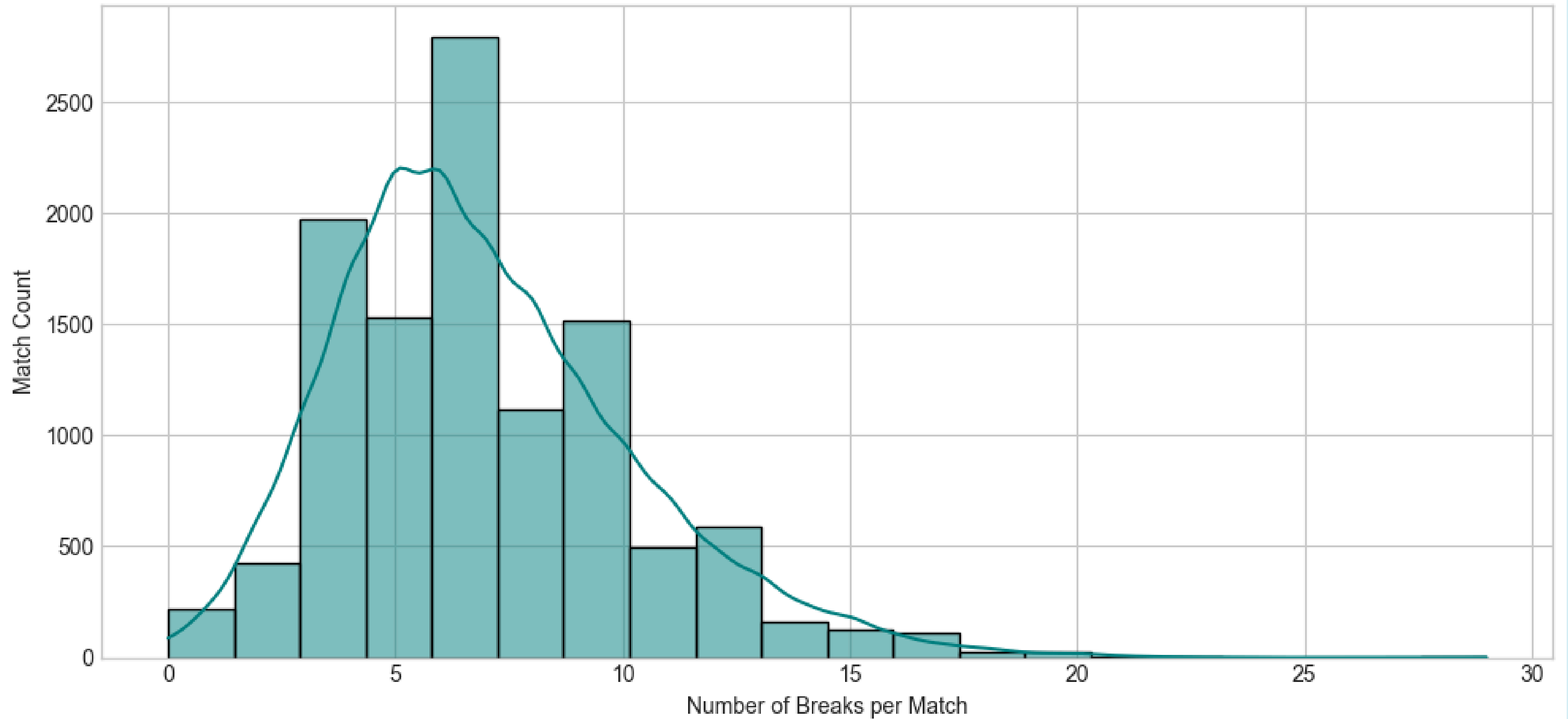


Question 17 :

What is the average number of breaks of serve per match?

Average Number of Breaks per Match: 6.93

Distribution of Breaks of Serve per Match



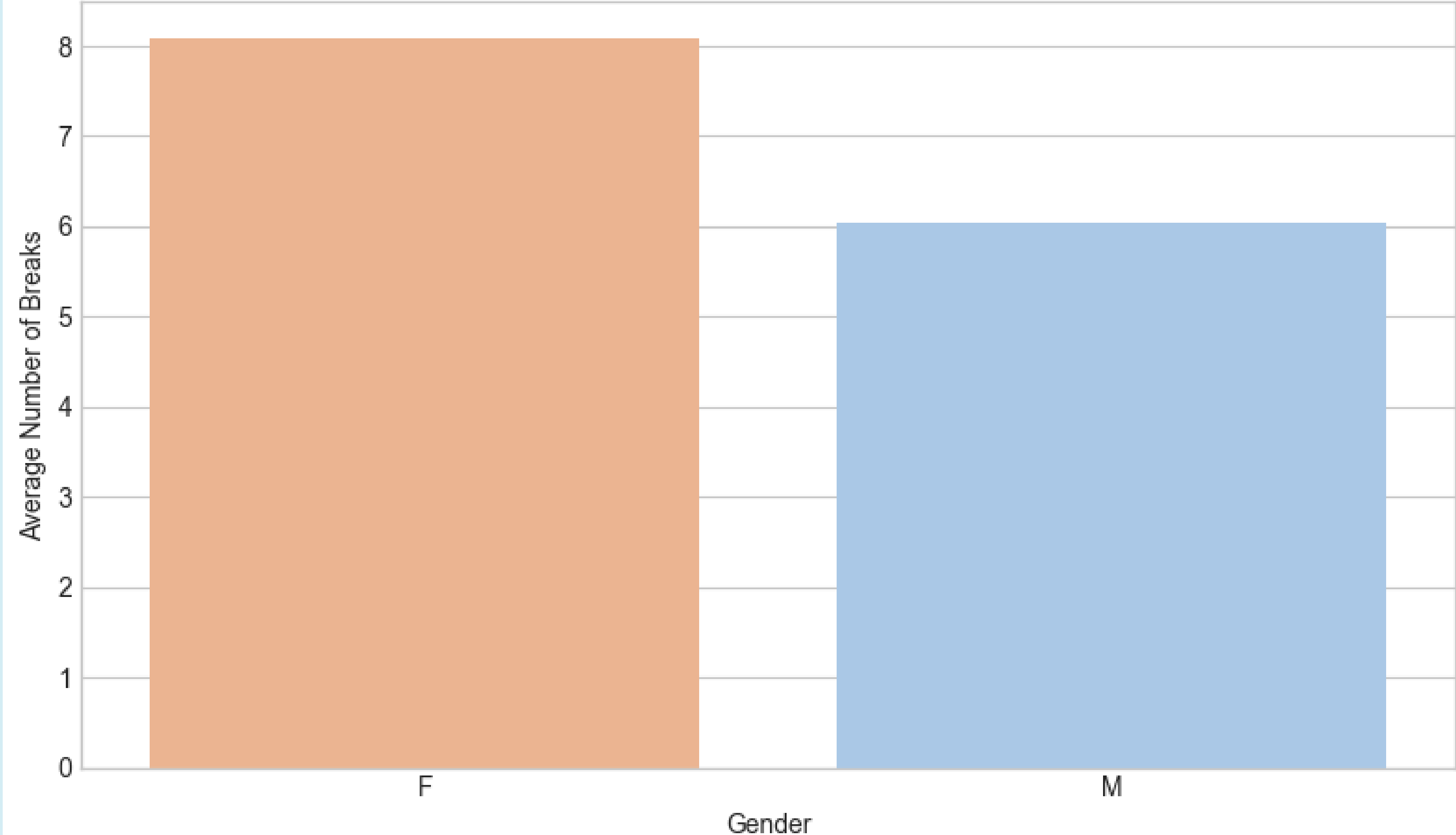
Average Number of Breaks per Match by Gender:

gender	num_breaks
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F	8.083681
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M	6.033898
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Average Breaks of Serve per Match by Gender



Thank you for your time.

link to the code:

https://github.com/marzi-dezfoulianrad/data_analysis_tennis